

Quantifying non-Gaussian information in the 21cm forest with information geometry

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We quantify how non-Gaussian structure contributes to constraints on X-ray heating and warm dark matter from the 21cm forest using information geometry. We compare the power spectrum (PS), wavelet scattering transform (WST), and topological data analysis (TDA), and assess phase dependence with random-phase surrogates that preserve the full Fourier amplitude spectrum. Within this information-geometric framework, the Fisher matrix defines a local metric on the (f_X, m_{WDM}) parameter manifold, and changes in its volume density quantify the loss of local model distinguishability under phase randomization. For simulated $z = 9$ spectra with SKA1-Low thermal noise corresponding to a 1000 h observation of a 10 mJy source, WST gives the strongest individual constraints at the benchmark under the standard regularized estimator. Its native Fisher-volume density is about 3.6 times that of the PS, and the surrogate WST remains substantially more informative than the PS. At this benchmark, phase randomization reduces the WST density by about one fifth to one third, depending on the estimator, and the loss remains positive across most of the parameter grid. Second-order coefficients show a larger fractional loss than first-order coefficients. TDA shows a larger fractional loss at the benchmark but substantially weaker absolute constraints. Thus the phase-sensitive component of non-Gaussian structure contributes to local parameter inference, while most of the WST Fisher-volume density in the main noisy analysis is reproduced after phase randomization. The relative importance of phase-sensitive structure is distinct from the overall constraining power of a summary.

I. INTRODUCTION

The period from recombination to the end of reionization spans the dark ages, Cosmic Dawn, and the Epoch of Reionization (EoR). The first stars, galaxies, and accreting compact objects altered the thermal and ionization state of the intergalactic medium (IGM), with reionization completing by $z \sim 5.5$ [1–3]. The redshifted HI 21 cm hyperfine line, with rest-frame frequency $\nu_{21} = 1420.4$ MHz, probes this evolution through absorption or emission relative to the radio background. Its brightness depends on the spin temperature, neutral fraction, density, and velocity field [1–5].

The 21cm forest complements measurements of the global 21 cm signal and its spatial fluctuations by using absorption features in the spectra of background radio sources. By analogy with the Ly- α forest, these features trace intervening neutral hydrogen, but through its hyperfine rather than electronic transition [6–12]. The feasibility of the measurement depends on the abundance, flux densities, and redshifts of compact radio-loud quasars and other active galactic nuclei [13–18]. The 21cm forest optical depth responds to the local neutral hydrogen density, spin temperature, and line-of-sight velocity gradient, making it sensitive to small-scale IGM structure [19–21].

Observations require bright high-redshift sources, high spectral resolution, and control of instrumental noise and calibration. Recent analyses of the radio-loud quasar

J352-15 nevertheless show that useful constraints are possible before a detection. Using archival uGMRT data, Šoltinský *et al.* [22] constrained the neutral fraction and temperature at $z \simeq 5.6$ from the one-dimensional 21cm forest power spectrum, disfavoring cold and substantially neutral models. Independently, Kongprachaya *et al.* [23] found no intervening absorption over $5.38 < z < 5.84$, constraining the optical depth and, under a residual-neutral-island model, the spin temperature. SKA1-Low will improve the sensitivity of such measurements [24, 25].

The 21cm forest is also sensitive to a broad range of small-scale cosmological physics. Previous studies have investigated its sensitivity to neutrino mass, running of the primordial spectral index, warm dark matter (WDM), ultralight dark matter, axion dark matter in post-inflationary Peccei–Quinn scenarios, the dark-matter–baryon relative velocity, primordial black holes, and oscillons [26–31]. Here we focus on the X-ray heating efficiency f_X and WDM mass m_{WDM} , which affect the absorption through the thermal state of the IGM and the suppression of small-scale structure, respectively. Their joint inference requires statistics that can distinguish these different physical effects.

Most forecasts and analyses use absorption-line counts, optical-depth distributions, or one-dimensional power spectra to describe the abundance, strength, and scale dependence of absorption [32–36]. The localized absorption structures of the 21cm forest can also carry non-Gaussian information beyond two-point statistics. Their

depths, widths, separations, and clustering reflect the thermal state of the IGM and its small-scale structure. The power spectrum describes the scale dependence of Fourier amplitudes, while relative Fourier phases also determine how fluctuations combine into the real-space absorption pattern. This motivates summaries that respond to the morphology of the absorption field and raises the question of how much that non-Gaussian structure improves physical parameter constraints.

The wavelet scattering transform (WST) provides deterministic and stable coefficients that describe fluctuations across scales. First-order coefficients summarize wavelet amplitudes, while second-order coefficients describe their modulation across broader scales [37, 38]. WST has been applied to large-scale structure, weak lensing, and galaxy clustering, as well as to high-redshift 21cm images and light cones [39–46]. Its application to the 21cm forest has also established sensitivity to X-ray heating and WDM free streaming [47].

Topological data analysis (TDA) provides another description of the absorption field through persistent homology, tracking the appearance and merger of connected components as a threshold is varied [48–51]. Related applications include the cosmic web, reionization topology, and cosmological forecasts, while 21cm forest TDA descriptors have been used to constrain heating and dark-matter free streaming [52–56]. Whereas WST describes wavelet amplitudes and their modulation, the TDA descriptors used here summarize the abundance and persistence of absorption components across thresholds. The two approaches therefore provide different descriptions of the same spectra.

Our central question is how much non-Gaussian structure contributes to the parameter information extracted from the 21cm forest. Establishing the origin of a summary’s constraining power requires more than comparing its performance with the power spectrum. A nonlinear summary can also use the detailed Fourier-amplitude information differently, so an improvement over the power spectrum alone leaves its dependence on non-Gaussian structure unresolved. Phase correlations provide a way to investigate this dependence because they encode aspects of nonlinear structure beyond two-point statistics [57]. Previous work compared WST coefficients from native and phase-randomized two-dimensional 21cm images to assess their non-Gaussian response [58]. Here we extend this comparison from changes in coefficients to changes in the ability to distinguish physical models.

Information geometry connects this question to local parameter inference by interpreting the Fisher matrix as a metric on the family of model distributions [59]. Related studies have assessed cosmological summaries from several complementary perspectives. Prelogović and Mesinger [60] compared cosmic 21cm summaries using the Fisher determinant, while Sui *et al.* [61] and Ref. [62] used mutual information to assess informativeness, sufficiency, and complementarity. Fisher-based Karhunen–Loève compression provides a classical ap-

proach to preserving parameter sensitivity in compressed data [63]. We use the Fisher-volume density $\sqrt{\det F}$ on the (f_X, m_{WDM}) parameter space to quantify joint local constraining power, together with marginalized errors to identify the effect on each parameter.

To assess the contribution of non-Gaussian structure through its phase dependence, we construct a random-phase surrogate for each noisy 21cm forest spectrum while preserving every Fourier amplitude. We then evaluate WST and TDA on the native and surrogate spectra using the same summary definitions. The change in their Fisher-volume densities provides an operational measure of the local parameter sensitivity lost under phase randomization. The native PS, WST, and TDA comparison establishes their relative constraining power, while the surrogate comparison measures how that constraining power depends on the original phase organization. We also examine the dependence on thermal noise and map second-order WST responses onto the physical wavelet and modulation scales to investigate how phase randomization changes the heating and WDM responses.

In the main noisy analysis, most of the WST Fisher-volume density remains after phase randomization, despite its improvement over the power spectrum. A smaller positive loss persists across most of the parameter grid, and second-order coefficients show a larger fractional loss than first-order coefficients. The reference-seed fiducial TDA comparison gives a moderately larger fractional loss, together with a smaller absolute Fisher volume and a less uniform grid response. These results quantify the usefulness of phase-dependent non-Gaussian structure for parameter inference and distinguish its relative importance from the overall constraining power of a summary.

II. BASICS OF THE 21 CM FOREST

The 21 cm forest arises from the absorption of radio emission from a background source by intervening neutral hydrogen. Along a direction $\hat{s}$ at redshift z , the 21 cm optical depth quantifies the strength of this absorption. In the optically thin limit, it is written as

$$\tau_{\nu_{21}}(\hat{s}, z) \simeq 0.0085 [1 + \delta(\hat{s}, z)] (1 + z)^{3/2} \times \frac{x_{\text{HI}}(\hat{s}, z)}{T_{\text{S}}(\hat{s}, z)} \left[\frac{H(z)/(1 + z)}{dv_{\parallel}/dr_{\parallel}} \right]. \quad (1)$$

Here δ is the local gas overdensity, x_{HI} is the neutral fraction, T_{S} is the spin temperature, $H(z)$ is the Hubble parameter, and $dv_{\parallel}/dr_{\parallel}$ is the velocity gradient along the line of sight. Equation (1) shows that the 21 cm forest is enhanced in cold, neutral, and overdense regions, whereas X-ray heating weakens the absorption by increasing the spin temperature.

The corresponding differential brightness temperature

relative to the background radiation is

$$\delta T_b(\hat{s}, \nu) \simeq \frac{T_S(\hat{s}, z) - T_\gamma(\hat{s}, \nu_{21}, z)}{1 + z} \tau_{\nu_{21}}(\hat{s}, z), \quad (2)$$

where $\nu_{21} = 1420.4$ MHz and T_γ denotes the brightness temperature of the background radio source at the redshifted 21 cm frequency. The observed 21cm forest spectrum is therefore determined by the density, neutral fraction, spin temperature, and line-of-sight velocity structure of the intervening IGM. In the simulations used here, the peculiar-velocity contribution to $dv_\parallel/dr_\parallel$ is neglected, so the velocity-gradient factor is evaluated using the Hubble expansion alone, consistently with Refs. [34, 35].

To model these physical dependencies, we use simulated line-of-sight 21 cm forest spectra generated with the multi-scale forward-modelling framework developed in Refs. [34, 35] and subsequently used in the WST & TDA analysis of Refs. [47, 56]. The method combines large-scale density and ionization fields with high-resolution modelling of the small-scale structures responsible for individual absorption features.

The large-scale density and ionization fields are generated with 21CMFAST [64–66] in a 1 cGpc box sampled by 500^3 cells, each 2 cMpc on a side. Neutral regions are identified in the large-scale volume, and the corresponding parent cells are refined to resolve the small-scale structures that dominate the 21cm forest signal.

Low-mass halos are populated according to a conditional halo mass function determined by the local large-scale overdensity and by the CDM or WDM matter power spectrum. The gas distribution is described by an NFW-based hydrostatic profile inside halo virial radii and by an infall profile outside them. The spin temperature is assumed to be coupled to the gas kinetic temperature through the Wouthuysen–Field effect. Gas inside halos is assigned the virial temperature, whereas gas outside halos follows an IGM thermal history including adiabatic expansion, Compton heating and cooling, and X-ray heating [34, 35].

We use the updated implementation of Sun *et al.* [35], in which each refined density field is matched to the overdensity of its parent 21CMFAST cell through an additive normalization rather than a multiplicative rescaling. The lower halo-mass cutoff is tied to the Jeans mass at the redshift of interest. The spectra analyzed in this work are evaluated at $z = 9$.

The present analysis varies two physical parameters that affect the 21cm forest in complementary ways. The X-ray heating efficiency f_X primarily controls the thermal state of the neutral IGM. Increasing f_X raises the gas and spin temperatures, suppressing deep absorption and reducing the contrast of the 21cm forest. The warm-dark-matter mass m_{WDM} instead regulates the abundance of small-scale structure through free-streaming suppression. Lower WDM masses suppress the formation of low-mass halos and modify the abundance and morphology of narrow absorption features. These contrasting responses motivate the joint consideration of f_X and m_{WDM} .

The model grid spans

$$m_{\text{WDM}} = 3, 4, 5, 6, 7, 8, 9 \text{ keV}, \quad (3)$$

and

$$f_X \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 1.0, 2.0, 3.0, 5.0\}. \quad (4)$$

For each grid model, the simulation ensemble contains ten neutral environments with different local mean densities and ten independent 10 cMpc line-of-sight realizations within each environment. We denote these by $N_{\text{dir}} = 10$ and $N_{\text{real}} = 10$, respectively. Thus each model contains 100 simulated spectra. The direction/environment grouping is retained throughout the statistical analysis so that resampling respects correlations among realizations drawn from the same large-scale environment. The 100 spectra should therefore not be interpreted as 100 independently observable high-redshift radio backlights.

We denote the resulting line-of-sight differential-brightness-temperature field by $\delta T_b(x)$, where the observed-frequency coordinate has been converted to a line-of-sight coordinate. The same simulated spectra are used as the common input to the power-spectrum, TDA, and WST analyses described in Sec. III.

To approximate observed spectra, we include thermal noise following the same flux-density radiometer prescription adopted in the 21cm forest TDA analysis [56]. The spectra are block-averaged in non-overlapping groups of four original samples, giving $N = 625$ channel centres from 2500 input samples, with no discarded samples. The actual centre spacing is $\Delta\nu_{\text{samp}} = 896.36$ Hz, or approximately 0.896 kHz. The thermal-noise prescription retains the nominal bandwidth $\Delta\nu_{\text{noise}} = 1$ kHz; this noise normalization is distinct from the sampling interval. The corresponding line-of-sight spacing used for the physical-scale mapping is $\Delta\chi = 0.0157187453$ cMpc, with centre-to-centre span 9.808497 cMpc and $N\Delta\chi = 9.824216$ cMpc. Gaussian, channel-independent thermal noise is then added to the continuum-normalized absorbed spectrum,

$$F_{\text{obs}}(\nu_i) = F(\nu_i) + q \sigma_{\text{rel}}(\nu_i) \xi_i, \quad \xi_i \sim \mathcal{N}(0, 1), \quad (5)$$

where $F(\nu) = \exp[-\tau(\nu)]$, σ_{rel} is the fiducial relative thermal-noise amplitude, and q is a dimensionless factor that rescales its strength. We take $q = 1$ for the fiducial 1000 h integration and $q = 0$ for the noiseless case; its dependence on integration time is specified below. In the optically thin limit, the corresponding rms optical-depth perturbation satisfies $|\delta\tau_\nu| \simeq |\delta F|/F$.

The rms flux-density noise per channel is

$$\sigma_S(\nu) = \frac{\text{SEFD}(\nu)}{\sqrt{n_{\text{pol}} t_{\text{int}} \Delta\nu_{\text{noise}}}}, \quad \text{SEFD}(\nu) = \frac{2k_B T_{\text{sys}}(\nu)}{A_{\text{eff}}(\nu)}. \quad (6)$$

Following Ref. [56], we adopt $n_{\text{pol}} = 2$ and a constant

SKA1-Low sensitivity ratio

$$\frac{A_{\text{eff}}}{T_{\text{sys}}} = 800 \text{ m}^2 \text{ K}^{-1}, \quad \text{SEFD} = \frac{2k_B}{A_{\text{eff}}/T_{\text{sys}}} \simeq 3.45 \text{ Jy}. \quad (7)$$

For the nominal noise bandwidth $\Delta\nu_{\text{noise}} = 1 \text{ kHz}$, this gives $\sigma_S \simeq 0.0407 \text{ mJy}$ at $t_{\text{int}} = 1000 \text{ h}$ and $\sigma_S \simeq 0.129 \text{ mJy}$ at $t_{\text{int}} = 100 \text{ h}$.

The flux-density noise is converted to a relative noise level using the background-source continuum,

$$\sigma_{\text{rel}}(\nu) = \frac{\sigma_S(\nu)}{S_\nu}, \quad S_\nu = S_{150} \left(\frac{\nu}{150 \text{ MHz}} \right)^\eta, \quad (8)$$

with fiducial values $S_{150} = 10 \text{ mJy}$ and $\eta = -1.05$. At 150 MHz , this corresponds to $\sigma_{\text{rel}} = 4.07 \times 10^{-3}$ for $t_{\text{int}} = 1000 \text{ h}$ and 1.29×10^{-2} for $t_{\text{int}} = 100 \text{ h}$. The noise model therefore corresponds to an SKA1-Low thermal-noise branch for a fiducial 10 mJy background source and is specified in relative-flux or optical-depth units rather than as a synthesized-beam brightness-temperature rms.

After the Gaussian thermal noise in Eq. (5) is added to the continuum-normalized 21cm forest spectrum, the resulting noisy spectrum $F_{\text{obs}}(\nu)$ is converted to the brightness-temperature convention of Eq. (2). PS, TDA, and WST are then evaluated on the same observed-like line-of-sight field, denoted $\delta T_{\text{b}}^{\text{obs}}(x)$. The main analysis adopts $t_{\text{int}} = 1000 \text{ h}$. To study the dependence on thermal-noise amplitude, we rescale the same fixed noise realization according to

$$q(t_{\text{int}}) = \left(\frac{1000 \text{ h}}{t_{\text{int}}} \right)^{1/2}, \quad (9)$$

with $q = 0$ denoting the noiseless case. The 3000 h , 1000 h , 300 h , and 100 h branches correspond to $q = 0.58, 1, 1.8$, and 3.2 , respectively. For each line of sight, we use the same Gaussian thermal-noise realization when evaluating neighboring points in the parameter grid. This ensures that differences between neighboring models arise from changes in the underlying 21cm forest signal rather than from independent noise fluctuations, thereby reducing noise in the numerical parameter derivatives.

III. SUMMARY STATISTICS

Before defining the statistics, we fix the terminology used throughout the paper. We call a quantity *native* when it is computed directly from the observed-like spectra used in the Fisher analysis, namely the observed-like 1 kHz spectra defined in Sec. II after the SKA1-Low $q = 1$ thermal-noise realization has been added and before any phase randomization. Thus the native power spectrum, native TDA, and native WST are all summaries of the same noisy line-of-sight fields $\delta T_{\text{b}}^{\text{obs}}(x)$. We call a quantity *surrogate* when it is computed after replacing each noisy native spectrum by a periodogram-preserving random-phase spectrum constructed from the

same line of sight. In particular, surrogate TDA and surrogate WST are not modified feature vectors; they are the TDA and WST summaries measured from the corresponding phase-randomized noisy surrogate spectra.

A. Native power spectrum

For each noisy 21cm forest spectrum $\delta T_{\text{b}}^{\text{obs}}(x)$, we subtract the line-of-sight mean and compute the one-dimensional periodogram. The resulting power is binned into 12 logarithmically spaced wavenumber bins.

This power spectrum is the two-point baseline for the same observed-like spectra used by WST and TDA. It has no explicit Fourier-phase sensitivity.

B. Wavelet scattering transform

We compute one-dimensional WST coefficients using Kymatio [38]. The input spectrum is convolved with a family of wavelets, modulus nonlinearities are applied, and the resulting amplitudes are averaged. Schematically, first-order coefficients have the form

$$S_1(j) = \langle |\delta T_{\text{b}} \star \psi_j| \rangle, \quad (10)$$

where ψ_j is a wavelet at scale j , $\star$ denotes convolution, and the angle brackets denote averaging along the line of sight. These coefficients measure the amount of localized structure at each wavelet scale.

Second-order coefficients apply a second wavelet transform to the first modulus field,

$$S_2(j_1, j_2) = \langle ||\delta T_{\text{b}} \star \psi_{j_1}| \star \psi_{j_2}| \rangle, \quad j_2 > j_1. \quad (11)$$

These coefficients measure how the amplitude of structure at the first wavelet scale is modulated at a larger scale. In the forest, such modulation can describe narrow absorption features embedded in broader absorption regions. Unlike a scale-by-scale amplitude description, it responds to their coherent organization, which is altered by phase randomization. Comparing the two WST orders therefore tests where the phase-sensitive parameter response is most prominent.

We compute WST coefficients up to second order using Kymatio with $J = 8$ and $(Q_1, Q_2) = (8, 1)$. We use the full 234-dimensional WST vector up to second order, including the zeroth-order coefficient, and also analyse the first- and second-order subsets separately. The magnitudes of the spatially averaged coefficients are logarithmically transformed for each sightline before ensemble averaging. Native and surrogate spectra are processed identically, without detrending or standardization.

For the physical-scale diagnostic, the wavelet centre frequencies are converted to comoving wavenumbers as

$$k_i = \frac{2\pi\xi_i}{\Delta\chi}, \quad i = 1, 2,$$

where $\Delta\chi$ is the comoving sampling interval. Here k_1 characterizes the scale selected by the first wavelet, whereas k_2 characterizes the larger-scale modulation of its modulus field.

C. Topological data analysis

For TDA, each spectrum is detrended with a smooth running mean and standardized to zero mean and unit variance. Writing the resulting field as $y(x)$, the one-dimensional sublevel-set filtration is

$$X_t = \{x \mid y(x) \leq t\}. \quad (12)$$

As the threshold t increases, new connected absorption components appear and existing components merge. In one dimension this evolution is summarized by a persistence diagram consisting of birth–death pairs (b_i, d_i) , where each pair records the threshold at which a connected component first appears and the threshold at which it merges into an older component. The corresponding lifetime is

$$\ell_i = d_i - b_i. \quad (13)$$

The TDA summaries therefore quantify the abundance, persistence, and arrangement of absorption troughs rather than only their Fourier amplitudes.

Following our previous 21cm forest TDA analysis, we retain only sufficiently persistent topological structures by imposing a lifetime threshold τ_* . We define the set of long-lived components as

$$I_{\text{long}} = \{i \mid \ell_i \geq \tau_*\}, \quad (14)$$

where ℓ_i is the persistence lifetime of the i th component.

The zeroth Betti number counts connected components. After the persistence cut, the Betti-0 curve is

$$\beta_0(t; \tau_*) = \#\{i \in I_{\text{long}} \mid b_i \leq t < d_i\}, \quad (15)$$

where b_i and d_i are the birth and death thresholds of the i th component, and $\#$ denotes the number of elements in the set. Thus, $\beta_0(t; \tau_*)$ gives the number of long-lived absorption components present at threshold t . The persistence cut suppresses short-lived structures associated with shallow or noise-dominated fluctuations. We adopt $\tau_* = 0.411$. The reference threshold t_* is fixed at the maximum of the mean Betti-0 curve of the noisy CDM baseline with $f_X = 0$ in the feature-generation pipeline, and the same threshold is used for all native and surrogate WDM models.

Following Ref. [56], we summarize the abundance and persistence of absorption components with the trough line density $\lambda(t_*)$, total squared persistence M_2 , and Betti-curve asymmetry A_{skew} . We also retain the fraction of finite persistence pairs surviving the lifetime cut, F_{long} , and their mean lifetime, $\tau_{\text{mean, long}}$.

The abundance and lifetime descriptors are

$$F_{\text{long}} = \frac{|I_{\text{long}}|}{N_{\text{pair}}}, \quad (16)$$

$$\tau_{\text{mean, long}} = \frac{1}{|I_{\text{long}}|} \sum_{i \in I_{\text{long}}} \ell_i, \quad (17)$$

$$M_2 = \sum_{i \in I_{\text{long}}} \ell_i^2, \quad (18)$$

$$\lambda(t_*) = \frac{\beta_0(t_*; \tau_*)}{L_{\parallel}}. \quad (19)$$

Here N_{pair} is the total number of finite persistence pairs in the diagram, and $L_{\parallel}$ is the line-of-sight extent of the spectrum.

The asymmetry of the Betti-0 curve is quantified from its cumulative area. Defining t_{05} and t_{95} as the thresholds that enclose 5% and 95% of the total area under $\beta_0(t; \tau_*)$, respectively, we define

$$A_{\text{skew}} = \frac{t_{95} - t_*}{t_* - t_{05}}. \quad (20)$$

This quantity measures the relative extent of the Betti-0 curve on the two sides of the reference threshold t_* .

The resulting TDA summary vector is

$$\mathbf{s}_{\text{TDA}} = \{\lambda(t_*), F_{\text{long}}, \tau_{\text{mean, long}}, M_2, A_{\text{skew}}\}. \quad (21)$$

These five descriptors provide a compact morphology-based summary of the abundance, persistence lifetime, and threshold-dependent organization of absorption structures in the noisy 21cm forest. Because the detrended input is standardized, they describe trough populations relative to the spectrum's own fluctuation amplitude rather than its absolute absorption depth. This differs from the scale-dependent amplitudes and modulations retained by the log-WST representation. Applying the same native–surrogate comparison to TDA tests whether the organization of persistent absorption components helps distinguish heating from WDM when the full Fourier-amplitude spectrum is held fixed.

IV. FULL-PERIDODOGRAM-PRESERVING RANDOM-PHASE SURROGATE SPECTRA

A. Construction

We construct phase-randomized surrogate spectra as controlled reference data, following the general logic of surrogate-data methods for nonlinear time series [67, 68]. We use this comparison to assess non-Gaussian parameter information through the dependence of a summary's sensitivity on the relative Fourier-phase organization of the 21cm forest.

For each native noisy spectrum $\delta T_{\text{b}}^{\text{obs}}(x)$, we subtract the line-of-sight mean and compute its real Fourier transform. For every independent nonzero Fourier mode, the

Fourier amplitude is kept fixed while the phase is replaced by an independent random draw,

$$\widetilde{\delta T}_{\text{b,surr}}(k) = \left| \widetilde{\delta T}_{\text{b}}^{\text{obs}}(k) \right| \exp(i\phi_k), \quad \phi_k \sim U(0, 2\pi). \quad (22)$$

The zero mode is left unchanged. For even-length spectra, the Nyquist mode is real and is treated separately. The inverse transform then produces a real-valued surrogate spectrum, after which the subtracted mean is restored. Because the Fourier amplitudes are unchanged, the native and surrogate spectra have the same line-of-sight periodogram by construction.

An absolute Fourier phase depends on the choice of spatial origin. A translation by x_0 changes the phases coherently as $\phi_k \rightarrow \phi_k - kx_0$ while leaving the morphology unchanged. The independent mode-by-mode randomization in Eq. (22) is different: it disrupts relative-phase couplings among Fourier modes and thereby reorganizes the localized absorption structure. Throughout this paper, “phase-sensitive” therefore refers to sensitivity to this relative Fourier-phase organization, or equivalently to morphology not fixed by the periodogram, rather than to the absolute phase of an individual mode.

For each surrogate seed, the same random-phase realization is used across the finite-difference stencil to reduce stochastic noise in the numerical parameter derivatives. To characterize the dependence on the particular random-phase realization, the WST surrogate analysis is repeated for eight phase seeds. Unless otherwise stated, surrogate-dependent scalar diagnostics are summarized by their seed median and 16th–84th percentile range. Each scalar quantity is aggregated independently across seeds, so median Fisher-volume elements, marginalized errors, and derived ratios need not correspond to a single common surrogate Fisher matrix.

The same WST and TDA procedures, including the same preprocessing and feature definitions, are applied to the native and surrogate spectra. Here “periodogram-preserving” means that the full line-of-sight periodogram of each noisy spectrum is preserved, rather than only the power-spectrum values used as the baseline. The surrogate WST and TDA statistics are therefore not equivalent to the power spectrum. They are computed from the phase-randomized spectra and can still respond to the detailed scale dependence encoded in the full set of Fourier amplitudes.

Figure 1 shows the same construction in real space for one fiducial sightline. Phase randomization rearranges the localized absorption pattern while preserving the periodogram to floating-point precision. This visualizes the distinction that underlies the native–surrogate comparison: the Fourier amplitudes are fixed, but the real-space organization is changed.

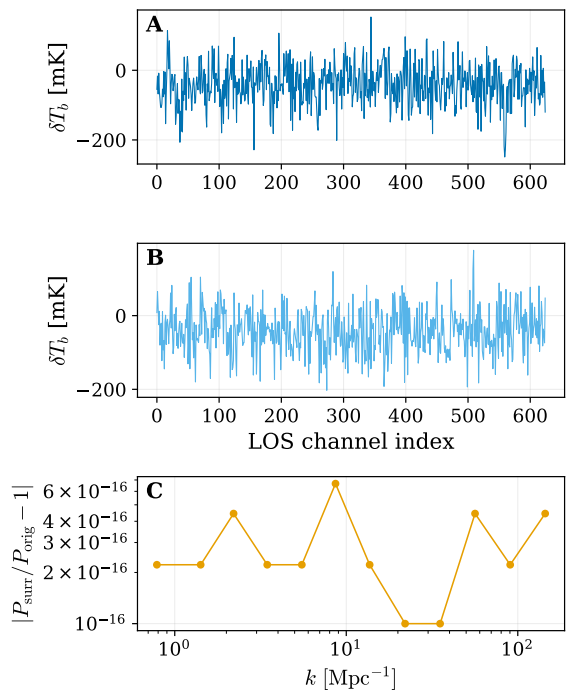


FIG. 1. A native noisy spectrum (top) and its random-phase surrogate (middle) at the fiducial model, using the main 1000 h thermal-noise branch normalized to a nominal 1 kHz bandwidth (Sec. II). Noise is added to the native field before its phases are randomized. The vertical axes show the converted brightness-temperature field δT_b^{obs} used by the summaries. The bottom panel shows the fractional power-spectrum difference, which is at floating-point precision despite the altered absorption pattern.

B. Operational non-Gaussian interpretation

For a statistically homogeneous Gaussian field with its mean specified, the power spectrum determines the field distribution and the independent Fourier modes have uniform random phases. Phase correlations therefore provide a way to examine non-Gaussian structure beyond two-point statistics. Our surrogate construction fixes the measured Fourier amplitudes of each 21cm forest spectrum and randomizes its original phase organization.

We assess the contribution of non-Gaussian structure to parameter inference through the change in WST and TDA sensitivity under this operation. A positive native-to-surrogate Fisher-volume loss indicates that the original phase organization improves the chosen summary’s joint local constraint. We interpret this as a phase-sensitive contribution of non-Gaussian structure under the adopted comparison. Sensitivity that remains after randomization is classified as *Fourier-amplitude-compatible*. It can differ from that of the PS because either nonlinear summary can respond to the detailed unbinned amplitudes and the real-space statistics they induce.

Phase randomization also changes the one-point dis-

tribution in general, so the comparison includes changes in absorption morphology beyond the positions of the troughs. The signed Fisher-volume diagnostics in Sec. V quantify the resulting change in local parameter sensitivity for each summary.

As a pipeline check, we apply the same native-surrogate WST analysis to Gaussian random-phase fields in Appendix A 1. The null fields do not show the coherent positive loss found for the 21cm forest.

V. FISHER INFORMATION GEOMETRY

A. Fisher metric from local distinguishability

In this work, we quantify information by asking how well a summary statistic can distinguish between nearby points in parameter space. Let $\mathbf{s}$ denote a summary vector and let $\boldsymbol{\theta} = (f_X, m_{\text{WDM}})$ denote the physical parameters. A choice of summary statistic defines a family of distributions $p(\mathbf{s}|\boldsymbol{\theta})$. The Kullback–Leibler divergence between neighboring models has the local expansion [69, 70]

$$D_{\text{KL}}[p(\mathbf{s}|\boldsymbol{\theta}) || p(\mathbf{s}|\boldsymbol{\theta} + d\boldsymbol{\theta})] = \frac{1}{2} F_{\alpha\beta} d\theta^\alpha d\theta^\beta + O(d\theta^3). \quad (23)$$

The Fisher matrix $F_{\alpha\beta}$ therefore acts as a metric tensor that measures local distinguishability in parameter space [71–73]. Different summary statistics induce different metrics on the same physical parameter space.

In practice, we adopt a Gaussian-summary approximation with the covariance held fixed at each fiducial model,

$$F_{\alpha\beta} = \frac{\partial \boldsymbol{\mu}^T}{\partial \theta^\alpha} \mathbf{C}_{\text{eff}}^{-1} \frac{\partial \boldsymbol{\mu}}{\partial \theta^\beta}, \quad (24)$$

where $\boldsymbol{\mu}(\boldsymbol{\theta})$ is the mean summary vector and $\mathbf{C}_{\text{eff}}$ is the regularized covariance evaluated at that model. This approximation retains the contribution from the parameter dependence of the mean and neglects covariance-derivative terms. We therefore interpret the resulting matrix as a local diagnostic of the mean-summary response rather than as a full likelihood forecast.

The parameter derivatives in Eq. (24) are evaluated numerically on the model grid defined in Sec. II. We restrict the calculation to the 45 interior grid points that have neighboring models on both sides of each parameter direction. For the uniformly spaced m_{WDM} grid, we use the standard three-point central difference. For the non-uniform f_X grid, we use the unequal-spacing three-point derivative obtained from a local quadratic interpolant.

Finite-ensemble noise in the estimated derivatives can bias the quadratic Fisher estimator high even after covariance regularization [74], and this bias need not cancel between the native and surrogate branches. We therefore perform an independent split-sample validation for WST, in which parameter-sensitive combinations derived from one subset of the direction/environment clusters are evaluated on a disjoint subset. This test asks whether

the native-to-surrogate loss persists when derivative noise is reduced. The detailed construction is given in Appendix A 2.

B. Information-volume element

For a positive-definite Fisher matrix in the two-parameter space, the local information-volume element is

$$dV_{\text{info}} = \sqrt{\det F} df_X dm_{\text{WDM}}. \quad (25)$$

This is the volume element associated with the Fisher metric and is invariant under smooth, invertible changes of parameter coordinates, as in the Jeffreys construction [75]. Here it serves as a local measure of joint parameter sensitivity.

Unlike the volume element, the density $\sqrt{\det F}$ depends on the parameter coordinates. It has inverse-parameter-area units, here keV^{-1} because f_X is dimensionless and m_{WDM} is expressed in keV. Native-to-surrogate ratios of this density at the same model are dimensionless and invariant under a common smooth, invertible change of parameter coordinates.

Within the local Gaussian approximation, the Fisher-ellipse area at a fixed contour level is proportional to $1/\sqrt{\det F}$. A larger density therefore corresponds to a smaller ellipse and stronger joint constraints. This measure accounts for parameter degeneracy, whereas a single marginalized error need not reflect the ordering of two-dimensional constraining power.

C. Native-to-surrogate information-volume loss

For a summary statistic S evaluated on both native and surrogate spectra, we compare their Fisher-volume densities at the same point in the (f_X, m_{WDM}) parameter space. The ratio

$$\frac{\sqrt{\det F_{S,\text{sur}}}}{\sqrt{\det F_{S,\text{native}}}} \quad (26)$$

measures how much of the native Fisher-volume density remains after phase randomization. We define the native-to-surrogate information-volume loss as the negative logarithm of this ratio,

$$\Delta I_{\text{vol}}^{(S)} = \log \frac{\sqrt{\det F_{S,\text{native}}}}{\sqrt{\det F_{S,\text{sur}}}} = -\log \frac{\sqrt{\det F_{S,\text{sur}}}}{\sqrt{\det F_{S,\text{native}}}}. \quad (27)$$

With this convention, a decrease in Fisher-volume density after phase randomization gives $\Delta I_{\text{vol}}^{(S)} > 0$, while no change gives zero. Since the Fisher-ellipse area at fixed contour level is proportional to $1/\sqrt{\det F}$, a positive $\Delta I_{\text{vol}}^{(S)}$ also means that the local constraint ellipse becomes larger after phase randomization.

For a more direct measure of the fractional change, we define

$$\eta_\phi^{(S)} = 1 - \frac{\sqrt{\det F_{S,\text{surr}}}}{\sqrt{\det F_{S,\text{native}}}} = 1 - \exp(-\Delta I_{\text{vol}}^{(S)}). \quad (28)$$

We refer to $\eta_\phi^{(S)}$ as the phase-sensitive Fisher-volume loss fraction. It quantifies the role of non-Gaussian structure through the phase-randomization comparison of Sec. IV B. A positive value means that this operation reduces the joint Fisher sensitivity of the chosen summary. For example, $\eta_\phi^{(S)} = 0.2$ means that the surrogate Fisher-volume density is 80% of the native value. In this case, 80% of the Fisher-volume sensitivity measured by that summary remains after the original phase organization has been removed, while 20% is lost under the surrogate operation.

The part that remains after phase randomization is classified here as Fourier-amplitude-compatible sensitivity. This should not be identified with the information contained in the PS used as our baseline. The surrogate construction preserves the full unbinned Fourier amplitudes of each noisy line of sight, whereas the native PS compresses them into 12 wavenumber bins. WST and TDA evaluated on the surrogate can therefore respond to detailed Fourier-amplitude structure that is absent from the binned PS. Likewise, $\eta_\phi^{(S)} = 0.2$ does not mean that 20% of the full likelihood information in the spectrum is non-Gaussian. The diagnostic measures the relative change in the chosen summary's Fisher-volume density under the specified comparison, rather than an additive component of its Fisher matrix.

If phase randomization removes morphology that carries useful parameter sensitivity, one naturally expects $\Delta I_{\text{vol}}^{(S)} > 0$ and $\eta_\phi^{(S)} > 0$. These signs are not imposed by construction. The native and surrogate summary distributions are not nested, and $F_{S,\text{native}} - F_{S,\text{surr}}$ is not assumed to be positive semidefinite. The loss measures can therefore become negative when the surrogate Fisher-volume density exceeds the native value. Such a case should not be interpreted as information being created by phase randomization. It only means that the chosen summary gives a larger local Fisher-volume density for the surrogate ensemble under the adopted diagnostic.

The volume-based diagnostics describe the joint constraint in the two-dimensional (f_X, m_{WDM}) parameter space, but they do not show how the uncertainty in each parameter changes. This distinction matters because phase randomization can change the size, shape, and orientation of the Fisher ellipse. An increase in ellipse area does not necessarily imply that the marginalized errors of both parameters increase by the same amount.

To determine how the native-surrogate difference appears in the individual parameter constraints, we define

the marginalized-error phase gain

$$\begin{aligned} G_\alpha^{(S)} &\equiv \frac{\sigma_{\alpha,S,\text{surr}}}{\sigma_{\alpha,S,\text{native}}}, \\ \sigma_{\alpha,S} &\equiv \sqrt{(F_S^{-1})_{\alpha\alpha}}, \\ \alpha &\in \{f_X, m_{\text{WDM}}\}. \end{aligned} \quad (29)$$

Here $\sigma_{\alpha,S}$ is the marginalized Fisher error on parameter α . Because the other parameter is allowed to vary, this error includes the effect of the degeneracy between f_X and m_{WDM} .

The ratio $G_\alpha^{(S)}$ measures how much the marginalized error changes after phase randomization. If phase-sensitive morphology helps constrain parameter α , removing that morphology increases the surrogate error and gives

$$G_\alpha^{(S)} > 1. \quad (30)$$

For example, $G_\alpha^{(S)} = 1.3$ means that the marginalized error from the surrogate is 30% larger than that from the native summary. A value $G_\alpha^{(S)} \simeq 1$ indicates little change. A value below unity means that the surrogate gives a smaller marginalized error under this diagnostic. Such a value does not by itself imply that Fourier-phase structure is detrimental to parameter inference.

If phase randomization removes useful information from the joint constraint and weakens both marginalized parameter constraints, the expected behavior is

$$\eta_\phi^{(S)} > 0, \quad G_{f_X}^{(S)} > 1, \quad G_{m_{\text{WDM}}}^{(S)} > 1. \quad (31)$$

However, $\eta_\phi^{(S)} > 0$ does not require both $G_{f_X}^{(S)}$ and $G_{m_{\text{WDM}}}^{(S)}$ to exceed unity. The Fisher ellipse can rotate or change shape as well as area. The two diagnostics therefore provide complementary information. The quantity $\eta_\phi^{(S)}$ measures the overall loss of joint Fisher sensitivity, while $G_\alpha^{(S)}$ shows how the change appears in each marginalized parameter constraint.

We evaluate these diagnostics for both WST and the five-descriptor TDA summary. The multi-seed and split-sample validation is carried out for WST. Unless stated otherwise, G_α below denotes the WST marginalized-error gain.

D. Covariance treatment

The covariance at each fiducial model is estimated from the local simulation ensemble. For the full WST summary, each spectrum provides 234 coefficients, whereas the ensemble contains $N = 100$ spectra. After mean subtraction, the sample covariance therefore has rank at most $N - 1 = 99$ and is singular, so it cannot be inverted directly for the Fisher calculation.

We regularize the covariance using the Schaefer–Strimmer diagonal-target shrinkage estimator [76],

$$\mathbf{C}_{\text{eff}} = (1 - \lambda)\mathbf{S} + \lambda\mathbf{T}, \quad \mathbf{T} = \text{diag}(\mathbf{S}), \quad (32)$$

where $\mathbf{S}$ is the sample covariance and λ is the shrinkage intensity estimated from the ensemble. This regularization preserves the individual feature variances while reducing uncertain off-diagonal covariances. The same prescription is applied to the native and surrogate summaries.

Appendix A3 gives the explicit covariance construction and examines the dependence of the WST native–surrogate loss on the surrogate phase seed and covariance estimator, including a comparison with Ledoit–Wolf shrinkage [77]. Finite-ensemble noise in the numerical parameter derivatives is a separate issue and is examined in Appendix A2.

OpenAI ChatGPT (GPT-5.6 Sol) was used to support the implementation and debugging of the analysis code, including code generation. The author provided the analysis specifications and reviewed and tested the resulting code before it was used in the calculations presented here.

VI. RESULTS

A. Constraining power and phase-sensitive non-Gaussian information

We first compare the absolute constraining power of PS, WST, and TDA, and then assess how non-Gaussian structure contributes through the phase dependence of each summary’s local parameter sensitivity. Figure 2 addresses the first question through the native PS, WST, TDA, and their joint constraint. Figure 3 addresses the second by comparing the native WST and TDA summaries with their random-phase counterparts.

At the benchmark point, the Fisher-volume densities are 607.3 for WST, 170.6 for PS, and 49.9 for TDA (Fig. 2 and Table I). WST therefore gives the strongest two-parameter constraint among the three individual summaries under the standard regularized estimator. The five TDA descriptors are evaluated on the same noisy spectra as PS and WST, following Ref. [56], which allows the cross-covariances among the three summaries to be estimated and incorporated into the joint analysis.

Combining PS, WST, and TDA gives $\sqrt{\det \mathbf{F}_{\text{joint}}} = 968.3$ (Fig. 2). Relative to WST alone under the same covariance treatment, this corresponds to a factor of 1.49 increase in Fisher-volume density. This increase is consistent with complementary parameter sensitivity among the three summaries within the adopted regularized estimator. Because the combined summary is high-dimensional, however, the numerical size of the gain is sensitive to covariance estimation, as examined in Appendix B.

Phase randomization reduces the WST Fisher-volume density but leaves most of it intact. For the surrogate

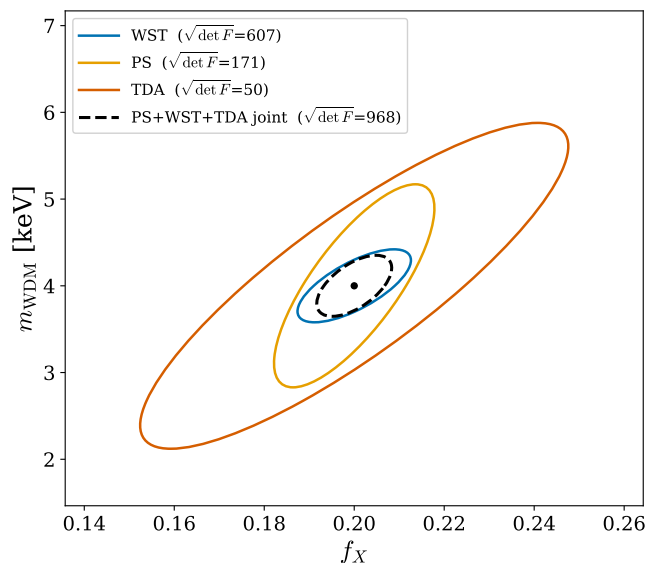


FIG. 2. Native PS, WST, TDA, and joint Fisher ellipses (68.3% regions) at $f_X = 0.2$ and $m_{\text{WDM}} = 4$ keV. All summaries are evaluated on the same 100 noisy spectra. The dashed joint ellipse uses the full covariance of the combined PS, WST, and TDA summary vector, including correlations among the three summaries. See Appendix B for details.

shown in Fig. 3, the Fisher-volume density decreases from 607.3 to 405.1 (Table I), corresponding to $\eta_{\phi}^{\text{WST}} \simeq 0.333$. Thus, about two thirds of the native WST Fisher-volume density remains after the original phase organization is removed. The surrogate WST also remains substantially more informative than the PS, indicating that much of the WST sensitivity is compatible with the preserved Fourier amplitudes.

TDA shows the same qualitative trend but a moderately larger fractional loss under phase randomization at this benchmark. Its Fisher-volume density decreases from 49.9 to 29.2 (Table I), corresponding to $\eta_{\phi}^{\text{TDA}} \simeq 0.415$. The Fisher ellipses in Fig. 3 show the associated degradation of the TDA constraint after phase randomization. The marginalized errors also increase and the parameter degeneracy becomes stronger. Thus, the native absorption morphology contributes appreciably to the TDA parameter sensitivity, although its absolute constraint remains weaker than those from PS and WST.

The two summaries therefore differ in both absolute constraining power and their fractional loss under phase randomization. For WST, most of the Fisher-volume density is reproduced after randomization, while the positive loss indicates additional sensitivity associated with the original absorption morphology. This comparison quantifies the phase-dependent contribution of non-Gaussian structure under the adopted surrogate construction. The moderately larger fractional loss of TDA at this benchmark is consistent with its sensitivity to the persistence and connectivity of absorption components,

TABLE I. Fisher constraints at $f_X = 0.2$ and $m_{\text{WDM}} = 4 \text{ keV}$. The surrogate rows correspond to the phase realization shown in Fig. 3. The units of $\sqrt{\det F}$ and $\sigma(m_{\text{WDM}})$ are keV^{-1} and keV , respectively.

Statistic	$\sqrt{\det F}$	$\sigma(f_X)$	$\sigma(m_{\text{WDM}})$	corr.
Native PS	170.6	0.0118	0.772	0.763
Native TDA	49.9	0.0314	1.239	0.857
Periodogram-preserving surrogate TDA	29.2	0.0527	2.522	0.966
Native WST	607.3	0.0083	0.277	0.700
Periodogram-preserving surrogate WST	405.1	0.0097	0.342	0.669

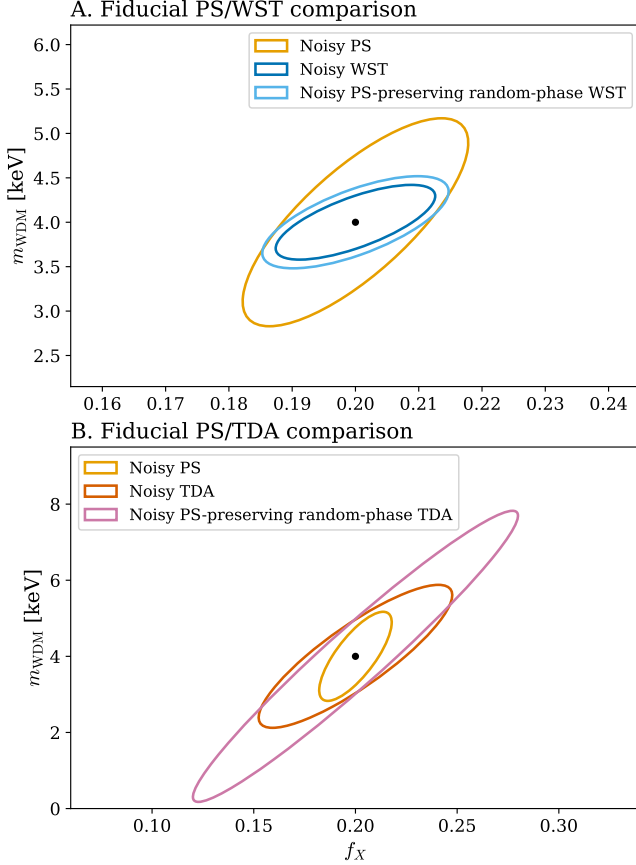


FIG. 3. Fiducial Fisher ellipses in the (f_X, m_{WDM}) plane. Top: native PS, native WST, and random-phase surrogate WST. Bottom: native PS, native TDA, and random-phase surrogate TDA. Most of the WST Fisher-volume density remains after phase randomization, while TDA shows a moderately larger fractional loss at this benchmark.

which change when the phases are randomized.

B. Thermal noise and the additional WST parameter gain

Panel A of Fig. 4 shows the Fisher-volume densities of PS, native WST, and surrogate WST as the thermal-noise amplitude is varied. All three decrease as the ther-

mal noise increases, with native WST remaining the most informative throughout the tested range and surrogate WST lying between native WST and PS. The native WST-to-PS Fisher-volume ratio changes from 24.2 in the noiseless case to 6.10 at 3000 h, 3.56 at 1000 h, and 2.78 at 300 h, before increasing to 5.92 at 100 h. Thus, although the absolute sensitivity of both summaries decreases toward higher thermal noise, their relative constraining power does not vary monotonically.

Panel B shows the fractional reduction of the WST Fisher-volume density under phase randomization. The eight-seed median η_{ϕ}^{WST} decreases from 0.505 in the noiseless case to 0.349 at 3000 h, 0.325 at 1000 h, and 0.137 at 300 h. This decrease means that the difference between native and phase-randomized WST becomes progressively smaller relative to the native WST constraint. In physical terms, increasing thermal noise obscures the phase-sensitive absorption morphology of the 21cm forest. As a result, the native and phase-randomized WST constraints become more similar, and the additional constraining power provided by the native phase organization is reduced.

At 100 h, the median loss increases again to 0.195. This high-noise upturn does not correspond to a recovery of absolute information: between 300 h and 100 h, the native WST Fisher-volume density decreases from 224.49 to 91.21, while the surrogate median decreases from 193.64 to 73.39. The surrogate therefore decreases slightly more strongly relative to the native WST, which increases their fractional separation. Moreover, the 16–84% intervals of η_{ϕ}^{WST} at 300 h (0.101–0.172) and 100 h (0.159–0.293) overlap. We therefore regard the high-noise increase as a modest non-monotonic feature of the native–surrogate comparison rather than as evidence that phase-sensitive information is restored by stronger thermal noise.

Panel C shows how the same comparison appears in the marginalized parameter constraints. At 1000 h, the eight-seed median gains are $G_{f_X} = 1.20$ and $G_{m_{\text{WDM}}} = 1.28$, corresponding to native errors about 17% and 22% smaller than the surrogate errors, respectively. The gain is larger for m_{WDM} than for f_X at every tested noise level. Toward stronger thermal noise, G_{f_X} decreases to 1.05 at 300 h and 1.01 at 100 h, so the phase-sensitive contribution to the marginalized f_X constraint becomes small. For m_{WDM} , the median gain decreases from 1.28 at 1000 h to 1.15 at 300 h, but then increases to 1.25 at

100 h. As in Panel B, this high-noise increase is a change in the relative native-surrogate constraint rather than a recovery of absolute information.

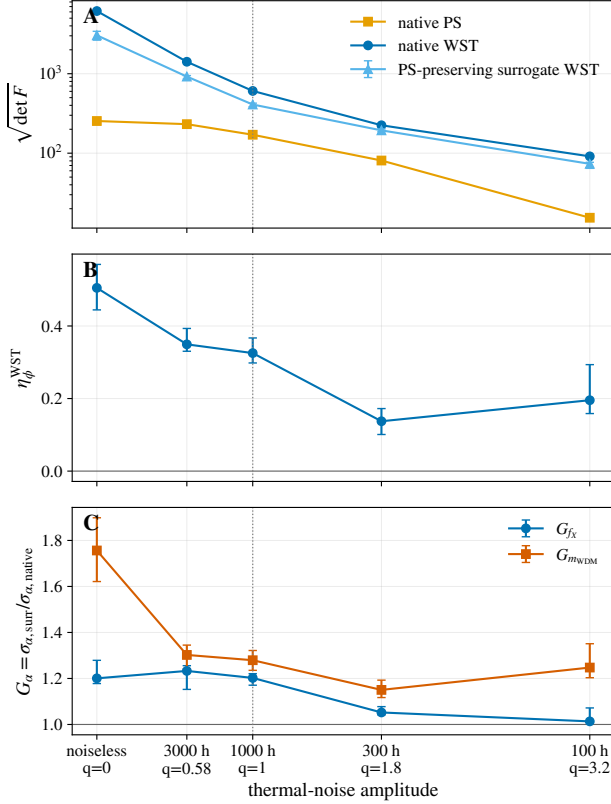


FIG. 4. Thermal-noise dependence of the Fisher constraints at $f_X = 0.2$ and $m_{\text{WDM}} = 4 \text{ keV}$. Panel A: Fisher-volume densities of PS, native WST, and random-phase surrogate WST. Panel B: fractional WST Fisher-volume loss η_ϕ^{WST} under phase randomization. Panel C: marginalized-error gains $G_\alpha = \sigma_{\alpha,\text{surr}}/\sigma_{\alpha,\text{native}}$ for f_X and m_{WDM} . The horizontal axis uses $q = (1000 \text{ h}/t_{\text{int}})^{1/2}$, with $q = 0$ denoting the noiseless case and the dotted line marking $q = 1$ (1000 h). The surrogate WST values in Panel A and the diagnostics in Panels B and C are shown as medians over eight phase seeds, with vertical bars indicating the corresponding 16–84% ranges.

C. Fiducial-grid behavior

The native-to-surrogate comparison was repeated at the 45 interior grid points where central finite differences are available. Figure 5 shows reference-seed maps of the fractional Fisher-volume loss η_ϕ for WST and TDA on the physical f_X – m_{WDM} grid.

For WST, the reference-seed loss is positive in all 45 interior cells, showing that the phase-sensitive contribution is not confined to the benchmark model.

For TDA, the reference-seed map is positive in 28 of 45 cells and shows sign changes across the grid, indicating

a less uniform dependence on the native phase organization. Negative values occur where phase randomization changes the parameter response and covariance of the TDA descriptors such that the surrogate summary yields a larger Fisher-volume density than the native summary. They therefore reflect a change in the Fisher sensitivity of the chosen summary, rather than the creation of new physical information by phase randomization.

The WST grid median obtained from all eight surrogate phase seeds is $\eta_\phi^{\text{WST}} = 0.349$ with the standard estimator (Table II). This value summarizes the seed median at each grid point, followed by the median across the grid. The table also reports the split-sample combined estimate, whose construction and validation are described in Appendix A 2.

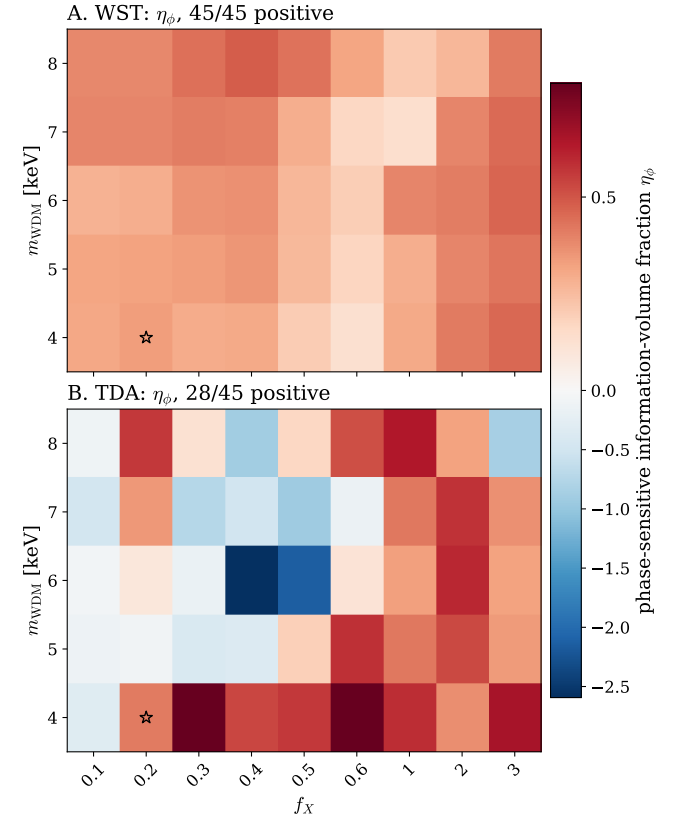


FIG. 5. Reference-seed maps of the fractional Fisher-volume loss η_ϕ under phase randomization over the f_X – m_{WDM} grid for the fiducial thermal-noise model. Top: WST. Bottom: TDA. Positive and negative values indicate a decrease and an increase, respectively, in the estimated Fisher-volume density after phase randomization. Negative values do not imply that phase randomization creates physical information. Both maps use the standard full-ensemble estimator and a single reference surrogate seed. The eight-seed WST grid summary is reported separately in Table II.

Panels (a) and (b) of Fig. 6 compare the absolute Fisher sensitivity of different WST orders with their fractional loss under phase randomization. Panel (a) shows the native and surrogate grid-median Fisher-volume den-

TABLE II. Finite-ensemble validation of the WST phase-sensitive loss. Both rows include all eight surrogate phase seeds. Square brackets give the fiducial 16–84% stability interval over seeds (standard) or paired split-seed estimates (combined). The grid column gives the median of the cell medians, with the number of positive cell medians in parentheses.

Estimator	Fiducial η_ϕ	Grid median
Standard	0.325 [0.298, 0.367]	0.349 (45/45)
Combined	0.200 [0.110, 0.282]	0.261 (43/45)

sities for the full WST and for the S_1 -only and S_2 -only subsets. Panel (b) shows the corresponding median and 16–84% range of the cell-by-cell fractional losses for the same reference seed used in Fig. 5.

For this reference seed, the S_1 -only and S_2 -only subsets have grid-median losses of 0.195 and 0.415, respectively. Thus, phase randomization removes a substantially larger fraction of the Fisher sensitivity carried by the second-order coefficients. This does not by itself imply that S_2 carries more absolute parameter information than S_1 . Rather, it shows that the constraining power of S_2 is more strongly reduced when the native phase organization is destroyed. Since second-order coefficients measure the modulation of first-order wavelet amplitudes across scales [Eq. (11)], the larger S_2 loss supports a connection between the phase-sensitive non-Gaussian contribution and the organization of absorption structure across scales.

Panels (c) and (d) show where phase randomization changes the parameter dependence of the second-order WST coefficients in wavelet-scale space. Let p denote a second-order WST path with wavelet scales (k_1, k_2) , and let $\mu_p^b(f_X, m_{\text{WDM}})$ denote the ensemble mean of that coefficient for branch $b \in \{\text{native}, \text{surr}\}$. At the fiducial model $(f_{X,0}, m_{\text{WDM},0}) = (0.2, 4 \text{ keV})$, we define

$$\begin{aligned}
 r_{p,f_X}^b &= f_{X,0} \frac{\partial \mu_p^b}{\partial f_X}, \\
 r_{p,m_{\text{WDM}}}^b &= m_{\text{WDM},0} \frac{\partial \mu_p^b}{\partial m_{\text{WDM}}}, \\
 \Delta r_{p,\alpha} &= r_{p,\alpha}^{\text{native}} - r_{p,\alpha}^{\text{surr}}, \quad \alpha \in \{f_X, m_{\text{WDM}}\}.
 \end{aligned} \tag{33}$$

The derivatives are evaluated using the same finite-difference prescription as in the Fisher analysis. The prefactors rescale the two parameter derivatives by their fiducial values, allowing their local responses to be compared on a dimensionless parameter scale. No covariance weighting is included in this diagnostic.

For the heating parameter, the native and surrogate responses are very similar at the higher modulation wavenumbers, whereas larger differences appear for some paths at lower k_2 . For example, at $(k_1, k_2) = (19.95, 2.186) \text{ cMpc}^{-1}$, $\Delta r_{f_X} = -0.123$.

The WDM response differences show a more complex scale dependence. Even at fixed $k_2 = 2.186 \text{ cMpc}^{-1}$,

the differences change sign with k_1 , reaching +0.149 and −0.169 for two neighboring paths. Thus, the effect of phase randomization on the second-order response is not associated with a single characteristic scale, but depends on the combination of first-wavelet and modulation scales. Panels (c) and (d) therefore provide a scale-resolved interpretation of the larger fractional loss of S_2 seen in panel (b). Supplementary native and surrogate response maps, together with a test of whether these differences persist after normalizing by the matching first-order amplitudes, are presented in Appendix C.

VII. SUMMARY AND DISCUSSION

We have used information geometry to quantify the contribution of phase-sensitive non-Gaussian structure to local parameter inference from the 21cm forest. The Fisher matrix defines the local distinguishability of models on the (f_X, m_{WDM}) parameter manifold, with $\sqrt{\det F}$ giving the associated local information-volume density. Phase randomization alters the morphology of the spectra while preserving their Fourier amplitudes, so that the native and surrogate summaries can be compared on the same physical parameter space. The fractional loss η_ϕ measures how much of the local parameter information carried by a summary is lost under phase randomization, as quantified by its Fisher-volume density.

For WST, most of the Fisher-volume density remains after phase randomization, although the loss is clearly nonzero. At the fiducial model, $\eta_\phi^{\text{WST}} = 0.325$ for the standard estimator and 0.200 for the split-sample combined estimator. The standard analysis also gives a positive loss throughout the parameter grid. Second-order coefficients lose a substantially larger fraction than first-order coefficients. TDA shows a somewhat larger fractional loss at the fiducial model, but its absolute Fisher-volume density is much smaller and its behavior across the parameter grid is less uniform.

The information-geometric interpretation is important here because the question is not simply whether the 21cm forest contains non-Gaussian structure. What matters for inference is whether that structure helps to distinguish nearby physical models. The fact that most of the WST Fisher sensitivity survives phase randomization shows that a substantial part of the response to X-ray heating and WDM is already present in structure compatible with the preserved Fourier amplitudes. The remaining loss reflects additional distinguishability associated with the native phase organization. In the main noisy analysis, this phase-sensitive contribution is measurable but smaller than the part that survives randomization.

The order decomposition gives a more direct physical interpretation of this additional contribution. Second-order WST coefficients describe how wavelet amplitudes at one scale are modulated on another, and their larger loss points to a stronger role for cross-scale organization

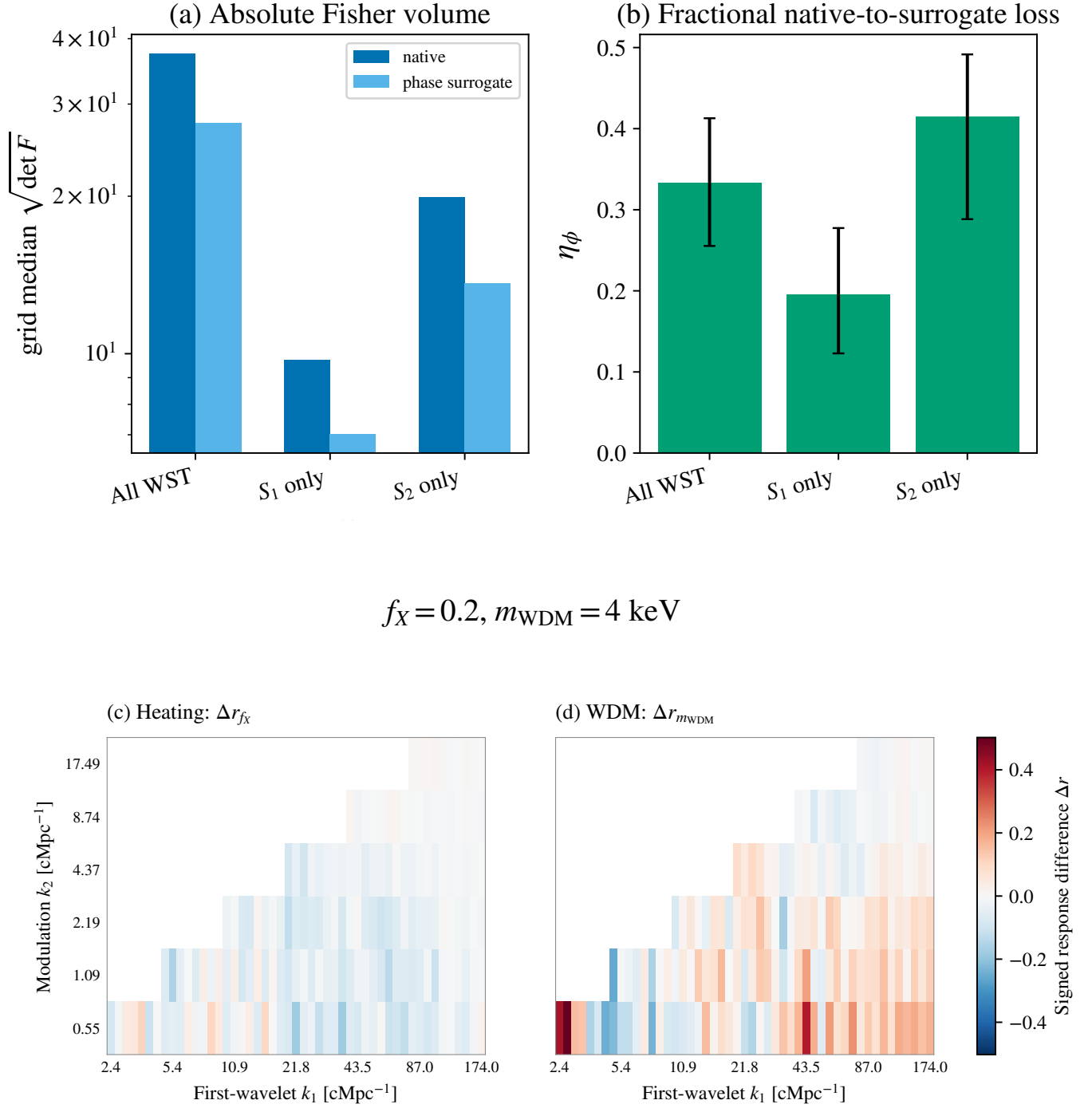


FIG. 6. WST order decomposition and scale-dependent parameter responses for the fiducial thermal-noise model. Panel (a) shows the grid-median native and surrogate Fisher-volume densities for the full WST and the S_1 -only and S_2 -only subsets. Panel (b) shows the corresponding fractional loss under phase randomization, with a larger loss for S_2 than for S_1 . Panels (c) and (d) show the signed native-surrogate differences in the second-order parameter responses for f_X and m_{WDM} , respectively, as functions of the wavelet scales (k_1, k_2) . Each cell represents an existing second-order WST path, while blank cells denote nonexistent paths.

than for single-scale amplitudes alone. Such behavior is consistent with the 21cm forest, where narrow absorption features occur within larger-scale thermal and density environments. The scale-pair response maps show that the native-surrogate differences depend on the particular combination of first-wavelet and modulation scales, rather than appearing uniformly across all second-order coefficients. Differences also remain after normalizing the second-order coefficients by the corresponding first-order amplitudes, so they cannot in general be explained by changes in first-order amplitude alone.

The TDA results show why fractional sensitivity to morphology and overall parameter constraining power should be kept separate. TDA loses a larger fraction of its Fisher-volume density under phase randomization than WST, while its absolute Fisher-volume density is much smaller. A summary statistic can therefore depend strongly on the native morphology in relative terms without providing correspondingly strong parameter constraints. WST, conversely, can constrain the parameters more strongly even though a smaller fraction of its sensitivity is phase dependent. In information-geometric terms, these are two different properties: the local Fisher-volume density itself and its fractional change under phase randomization.

The interpretation of η_ϕ depends on the specific surrogate construction used here. Phase randomization preserves the Fourier amplitudes but changes the phase organization and, in general, the one-point distribution. It therefore does not provide a complete decomposition of Gaussian and non-Gaussian information. Instead, η_ϕ measures the loss of Fisher sensitivity produced by this specific transformation. The result is local in parameter space and also depends on the adopted summary statistics and observational model. The larger WST loss in the noiseless case suggests that thermal noise reduces the relative visibility of phase-sensitive structure, although the dependence on noise level is not strictly monotonic. A survey-level forecast will additionally require instrumental systematics and a model for the observable background radio-source population.

VIII. CONCLUSIONS

We have used information geometry to quantify the contribution of phase-sensitive non-Gaussian structure to parameter inference from the 21cm forest. By comparing the local Fisher geometry of native and phase-randomized spectra, we find that most of the WST sen-

sitivity survives phase randomization, while a measurable loss remains. The strong WST constraints therefore arise predominantly from information that remains compatible with the preserved Fourier amplitudes, with an additional contribution from the native phase organization.

The phase-sensitive contribution is more pronounced in second-order than in first-order WST coefficients, linking it to the cross-scale organization of the absorption structure. TDA shows a stronger fractional dependence on phase randomization but substantially weaker absolute constraining power, demonstrating that sensitivity to non-Gaussian morphology and overall parameter information need not track each other. These results show how an information-geometric comparison can separate the overall constraining power of a summary from the contribution associated with phase-sensitive non-Gaussian structure, and provide a framework for identifying which aspects of 21cm forest morphology carry useful cosmological information.

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DATA AND CODE AVAILABILITY

The simulated 21cm forest spectra analyzed in this work were generated with the forward-modelling framework described in Refs. [34, 35]. The derived data products and analysis scripts required to reproduce the results and figures presented in this paper will be made available in a public repository upon publication. The underlying simulation data are available from the corresponding author upon reasonable request.

Appendix A: Robustness tests

1. Gaussian-power null validation

As a diagnostic check of the surrogate-Fisher pipeline, we repeat the WST native-surrogate comparison on Gaussian-power null fields. For each point in the parameter grid, the target one-dimensional power spectrum is

estimated from the corresponding noisy spectra and used to generate Gaussian random-phase fields with the same spectrum length and preprocessing conventions as the main analysis. These null fields retain parameter-dependent Fourier-amplitude information but contain no coherent phase-dependent absorption morphology. We then apply the same full log-WST feature extraction, covariance regularization, non-uniform finite-difference derivatives, surrogate construction, and Fisher-volume calculation used for the 21cm forest spectra.

Figure 7 shows that the null values of η_ϕ^{WST} are centered close to zero and do not reproduce the coherent positive WST signal measured from the 21cm forest grid. Over the 45 interior fiducial models, the null median is 0.006, the mean is 0.011, and the 16–84% interval is -0.095 – 0.123 . Of the 45 cells, 24 are positive, close to an even split. The median ratio of the surrogate to native Fisher-volume elements is $\sqrt{\det F_{\text{surr}}}/\sqrt{\det F_{\text{native}}} = 0.994$, so the native and surrogate null Fisher volumes are also close to parity at the median level.

We use this test only as a pipeline sanity check. It is not a formal null distribution or significance calibration for the 21cm forest result. Rather, it verifies that the analysis does not automatically produce a large coherent positive native–surrogate WST loss when coherent phase-dependent absorption morphology is absent.

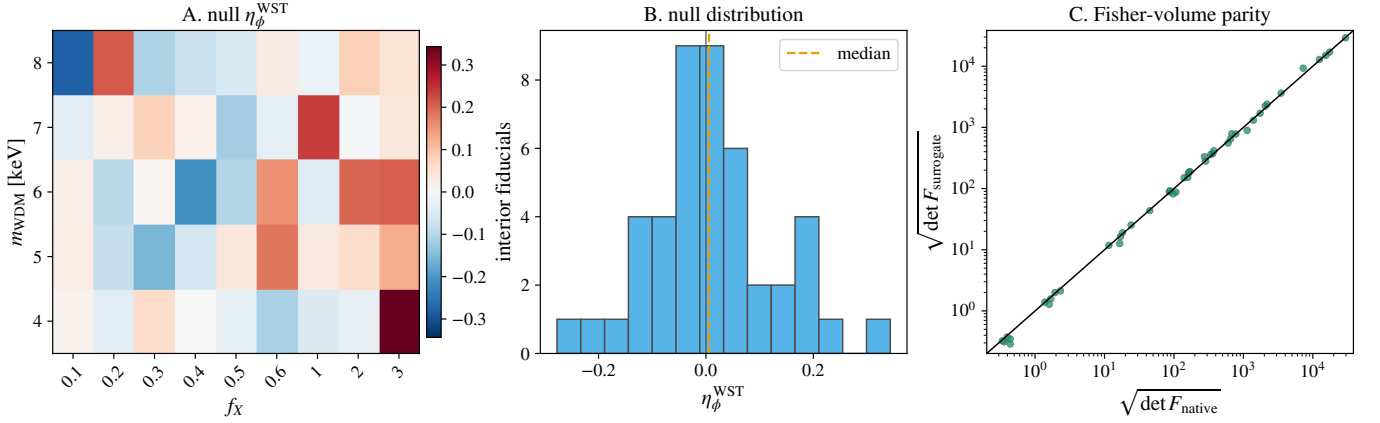


FIG. 7. Gaussian-power null validation of the WST native–surrogate Fisher-volume loss. Panel A shows η_ϕ^{WST} over the interior grid, panel B its distribution, and panel C the native and surrogate Fisher-volume elements. The null distribution is centered close to zero, and the native and surrogate Fisher volumes are close to parity. This test is used as a pipeline sanity check rather than as a formal null distribution or significance calibration for the 21cm forest result.

2. Finite-ensemble Fisher validation

Finite-sample noise in the numerical parameter derivatives can bias the quadratic Fisher estimator high. We test this effect with a split-sample score-compression analysis following Ref. [74]. The ten direction/environment clusters are divided into a compression set A and a disjoint evaluation set B , while the ten realizations belonging to each direction are kept together. We use 5/5, 6/4, and 7/3 splits. All 582 possible partitions are used at the fiducial model, and 100 fixed partitions are used at each interior grid point.

For each native or surrogate sample, set A is used to estimate the mean μ_A , parameter derivatives $\mathbf{D}_A$, and regularized covariance $\mathbf{C}_A$. These define the two-dimensional compression

$$\mathbf{t}_A(\mathbf{s}) = \mathbf{D}_A^T \mathbf{C}_A^{-1} (\mathbf{s} - \mu_A). \quad (\text{A1})$$

The compression is then held fixed and evaluated on the independent set B , giving F_{comp} . We combine this with the standard Fisher matrix obtained from set A through the positive-definite matrix geometric mean

$$F_{\text{comb}} = F_{\text{std},A}^{1/2} \left(F_{\text{std},A}^{-1/2} F_{\text{comp}} F_{\text{std},A}^{-1/2} \right)^{1/2} F_{\text{std},A}^{1/2}. \quad (\text{A2})$$

Native WST and each surrogate seed are treated separately, and η_ϕ^{WST} is evaluated for each matched split–seed pair before aggregation. Because the partitions overlap, the quoted percentile ranges measure empirical stability rather than statistical confidence.

The split-sample test reduces the inferred loss but preserves its positive sign. At the fiducial model, the eight-seed median decreases from 0.325 with the standard estimator to 0.200 with the combined estimator. Across the parameter

grid, the corresponding medians are 0.349 and 0.261, with positive cell medians in 45/45 and 43/45 cells, respectively. At the fiducial model, 97.8% of the matched split-seed estimates remain positive.

Further tests give the same qualitative result. Deleting one direction at a time gives combined fiducial medians between 0.133 and 0.283, and alternative finite-difference stencils preserve a positive fiducial loss. The larger fractional loss of the second-order WST subset relative to the first-order subset is also retained. These tests show that finite-ensemble effects influence the magnitude of η_ϕ^{WST} but do not account for its positive sign or for the stronger loss of the second-order coefficients.

3. Surrogate phase seeds and covariance estimators

We give further details of the covariance regularization and test whether the WST native-surrogate fractional loss depends on the surrogate phase seed or on the covariance estimator. Figure 8 summarizes the latter two tests using the standard full-ensemble Fisher estimator.

Let $\mathbf{s}^{(r)}$ denote the p -dimensional summary vector for realization r , treated as a column vector, and let $\bar{\mathbf{s}}$ be the ensemble mean. Defining the centered vector

$$\mathbf{x}^{(r)} = \mathbf{s}^{(r)} - \bar{\mathbf{s}}, \quad (\text{A3})$$

the sample covariance is

$$\mathbf{S} = \frac{1}{N-1} \sum_{r=1}^N \mathbf{x}^{(r)} \mathbf{x}^{(r)\text{T}}. \quad (\text{A4})$$

Equivalently, collecting the centered realizations as rows of the $N \times p$ matrix

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)\text{T}} \\ \mathbf{x}^{(2)\text{T}} \\ \vdots \\ \mathbf{x}^{(N)\text{T}} \end{pmatrix}, \quad (\text{A5})$$

gives

$$\mathbf{S} = \frac{1}{N-1} \mathbf{X}^{\text{T}} \mathbf{X}. \quad (\text{A6})$$

Because the centered realizations satisfy

$$\sum_{r=1}^N \mathbf{x}^{(r)} = 0, \quad (\text{A7})$$

at most $N-1$ rows of $\mathbf{X}$ are linearly independent. Hence

$$\text{rank}(\mathbf{S}) = \text{rank}(\mathbf{X}) \leq N-1. \quad (\text{A8})$$

For the full WST, the number of summary coefficients exceeds the number of simulated realizations, placing the covariance estimation in a high-dimensional, low-sample-size regime also encountered in astronomical data analysis [78]. With $p = 234$ and $N = 100$,

$$\text{rank}(\mathbf{S}) \leq 99 < 234, \quad (\text{A9})$$

so the raw sample covariance is singular and cannot be inverted directly.

We use the shrinkage estimator defined in Eq. (32). Our default choice is the Schaefer–Strimmer diagonal-target estimator [76], for which $\mathbf{T} = \text{diag}(\mathbf{S})$. The shrinkage intensity is estimated from the sampling variability of the off-diagonal covariance elements following Ref. [76]. Since the target has the same diagonal as the sample covariance,

$$(C_{\text{eff}})_{ii} = S_{ii}, \quad (C_{\text{eff}})_{ij} = (1 - \lambda) S_{ij} \quad (i \neq j). \quad (\text{A10})$$

The individual feature variances are therefore unchanged, while poorly determined off-diagonal covariances are shrunk toward zero. This produces a regularized covariance that can be inverted stably for the Fisher calculation. The same estimation and regularization procedure is applied separately to the native and surrogate summary ensembles.

The left panel of Fig. 8 shows η_ϕ^{WST} for eight surrogate phase seeds using the default Schaefer–Strimmer estimator. All eight seeds give a positive loss, with a median of $\eta_\phi^{\text{WST}} = 0.325$ and a 16–84% range of 0.298–0.367. The positive loss is therefore not specific to a particular random-phase realization.

The right panel tests the dependence on covariance regularization by repeating the analysis with the Ledoit–Wolf shrinkage estimator [77]. Both Schaefer–Strimmer and Ledoit–Wolf give a positive seed-median loss, although the inferred magnitude changes with the covariance estimator. The positive sign of the WST loss is therefore not specific to the default covariance regularization.

These tests address the dependence on surrogate phase realization and covariance estimation. Finite-ensemble noise in the numerical parameter derivatives is tested separately in Appendix A 2.

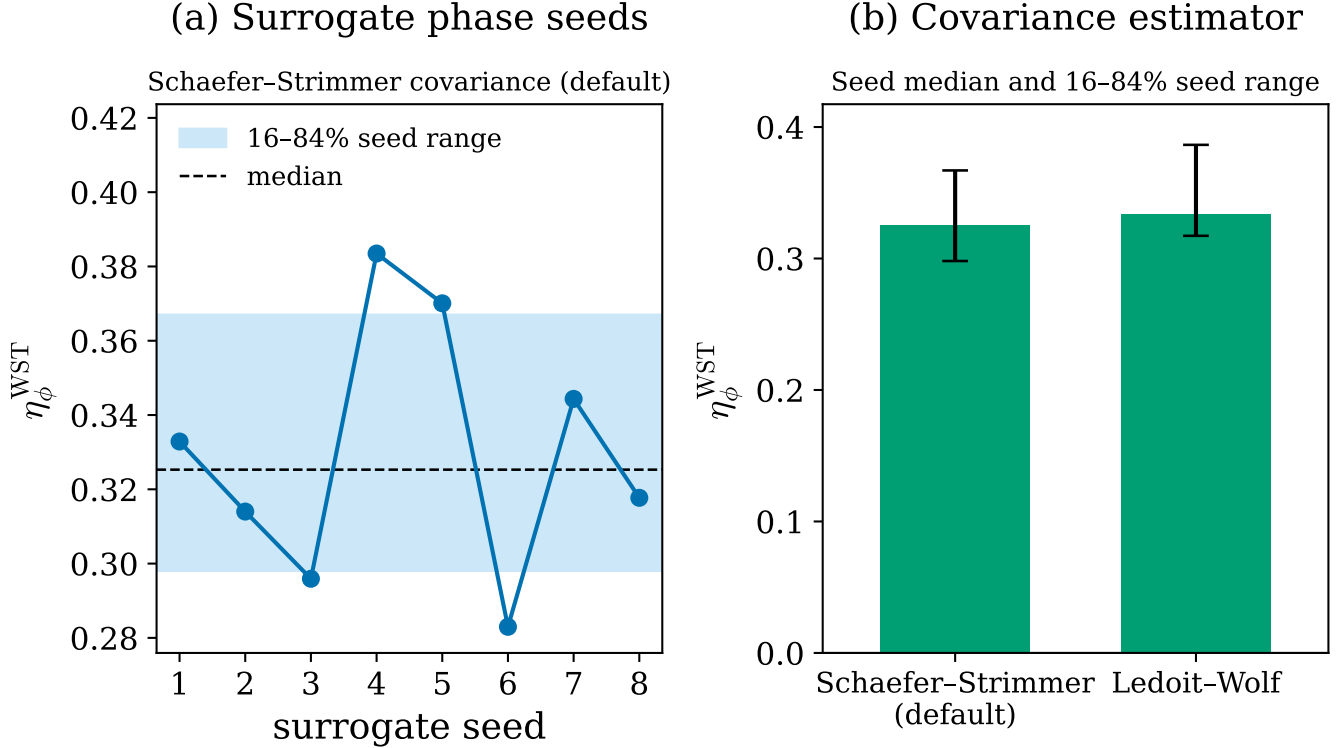


FIG. 8. Robustness of the WST fractional loss under phase randomization. Left: η_ϕ^{WST} for eight surrogate phase seeds using the default Schaefer–Strimmer covariance estimator. The horizontal line and shaded band show the median and 16–84% range across seeds. Right: the seed-median η_ϕ^{WST} obtained with the default Schaefer–Strimmer estimator and the Ledoit–Wolf alternative, with vertical bars showing the corresponding 16–84% seed ranges. The median loss remains positive for both covariance estimators. The displayed ranges describe variation across surrogate phase seeds, not confidence intervals.

Appendix B: Joint-summary covariance and pairing diagnostics

For the joint PS–WST–TDA analysis, the three summaries are matched by simulation model, direction, and line of sight before concatenation, so that all features are measured from the same noisy 21cm forest spectrum. The concatenated vector contains 12 PS, 234 WST, and 5 TDA features, for a total dimension of 251. Its covariance includes all cross-summary covariance blocks and is regularized with the Schaefer–Strimmer shrinkage estimator [76]. At the benchmark, the joint Fisher-volume density is 968.3 keV^{-1} . Extracting the WST block from the same regularized covariance gives 651.5 keV^{-1} , corresponding to a nested joint-to-WST ratio of 1.49.

Because the number of features exceeds the number of simulated spectra, the joint inverse covariance is sensitive to regularization. The sample correlation matrix has rank 99, and regularization-dominated directions make a substantial

contribution to the estimated Fisher matrix. We therefore test the dependence of the joint result on both the observed cross-summary pairing and the covariance estimator.

To isolate the role of cross-summary covariance, we permute the PS and TDA rows relative to WST while preserving the marginal summary samples. Both global and within-direction permutations give larger joint Fisher volumes on average than the observed matched pairing, with a smaller shift for the within-direction case. The result is qualitatively unchanged when the shrinkage intensity is either fixed or re-estimated.

A separate Gaussian calibration with a known block-diagonal population covariance gives a population joint-to-sum ratio of unity, while the same finite-sample regularized estimator gives a median ratio of 1.22. This shows that covariance estimation alone can inflate high-dimensional Fisher-volume ratios. Additional delete-direction, split-sample, and lower-dimensional tests likewise show that the numerical size of the joint gain depends on the representation and estimator. We therefore do not interpret the factor of 1.49 as a direct measure of independent physical information contributed by the added summaries.

Appendix C: Supplementary WST response diagnostics

This appendix examines where the native–surrogate differences in the second-order WST appear in wavelet-scale space and whether they can be explained solely by changes in the corresponding first-order amplitudes. The coordinates (k_1, k_2) denote the characteristic wavenumbers of the wavelet filters and should not be interpreted directly as absorber sizes.

Figure 9 compares the native and surrogate second-order parameter responses defined in Eq. (33). The overall scale dependence is similar between the two cases, particularly for f_X , while the native–surrogate differences vary across the (k_1, k_2) plane. The maps therefore show that the response to phase randomization is not uniform across the second-order WST coefficients.

We next test whether these second-order differences can be attributed solely to changes in the corresponding first-order amplitudes. For each sightline ℓ and second-order path p , we define

$$R_{21,p,\ell}^b = \frac{S_{2,p,\ell}^b}{S_{1,n_1(p),\ell}^b}, \quad b \in \{\text{native, surr}\}, \quad (\text{C1})$$

where $n_1(p)$ denotes the first-order coefficient associated with the first wavelet of the second-order path p . The ratio is formed sightline by sightline before ensemble averaging.

The Fisher analysis uses logarithmically transformed WST coefficients. For this normalization test, the corresponding positive coefficient magnitudes are reconstructed by inverting the logarithmic transformation before forming R_{21} . We then define the ensemble mean

$$\mu_p^{R,b} = \langle \ln R_{21,p,\ell}^b \rangle_\ell \quad (\text{C2})$$

and the native–surrogate difference in its scaled parameter response,

$$\Delta r_{p,\alpha}^{R_{21}} = a_\alpha \frac{\partial}{\partial \theta^\alpha} (\mu_p^{R,\text{native}} - \mu_p^{R,\text{surr}}). \quad (\text{C3})$$

The parameter scaling a_α is the same as in Eq. (33).

Figure 10 shows that nonzero native–surrogate response differences remain after the S_2/S_1 normalization. The second-order differences therefore cannot in general be reduced to changes in the matching first-order amplitudes alone. This normalization is used only as a coefficient-level diagnostic. All Fisher matrices and parameter constraints reported in the main analysis use the original log-WST summary.

Second-order log-WST: native and surrogate

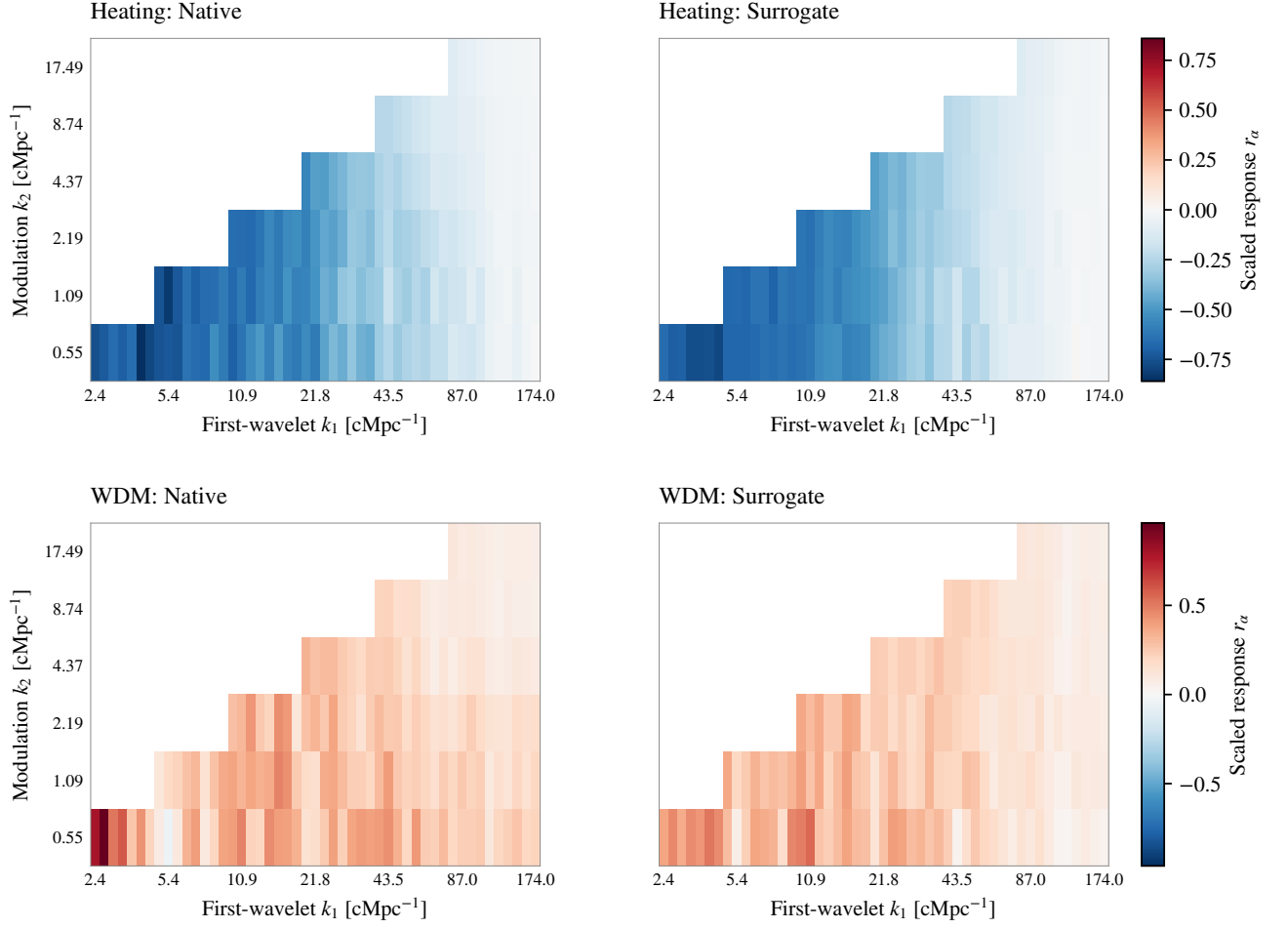


FIG. 9. Native and surrogate second-order WST parameter responses for f_X (top) and m_{WDM} (bottom). Each cell represents an existing second-order path at wavelet-filter centres (k_1, k_2) . Native and surrogate maps share the same symmetric colour scale within each parameter row. The response differences vary across wavelet-scale pairs rather than appearing uniformly over the second-order coefficients. Blank cells denote nonexistent paths.

Amplitude-normalized second order: signed response differences

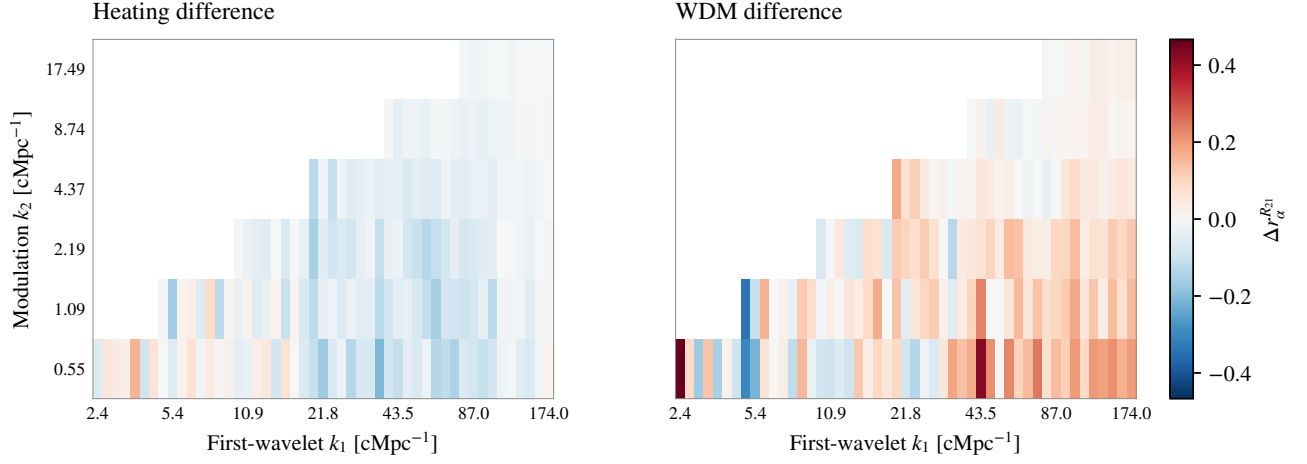


FIG. 10. Native-surrogate differences in the second-order parameter response after normalizing each second-order coefficient by its matching first-order coefficient, for f_X (left) and m_{WDM} (right). Each cell represents an existing second-order WST path at wavelet-filter centres (k_1, k_2) . Nonzero differences after the S_2/S_1 normalization show that the scale-dependent response differences cannot in general be reduced to changes in the corresponding first-order amplitudes alone. Blank cells denote nonexistent paths.

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- [1] S. R. Furlanetto, S. P. Oh, and F. H. Briggs, *Physics Reports* **433**, 181 (2006), arXiv:astro-ph/0608032.
- [2] J. R. Pritchard and A. Loeb, *Reports on Progress in Physics* **75**, 086901 (2012), arXiv:1109.6012 [astro-ph.CO].
- [3] H. Shimabukuro, K. Hasegawa, A. Kuchinomachi, H. Yajima, and S. Yoshiura, *Publications of the Astronomical Society of Japan* **75**, S1 (2023), arXiv:2303.07594 [astro-ph.CO].
- [4] D. Scott and M. J. Rees, *Monthly Notices of the Royal Astronomical Society* **247**, 510 (1990).
- [5] P. Madau, A. Meiksin, and M. J. Rees, *The Astrophysical Journal* **475**, 429 (1997), arXiv:astro-ph/9608010.
- [6] C. L. Carilli, N. Y. Gnedin, and F. Owen, *The Astrophysical Journal* **577**, 22 (2002), arXiv:astro-ph/0205169.
- [7] C. L. Carilli, N. Gnedin, S. Furlanetto, and F. Owen, *New Astronomy Reviews* **48**, 1053 (2004), arXiv:astro-ph/0409311.
- [8] S. R. Furlanetto and A. Loeb, *The Astrophysical Journal* **579**, 1 (2002), arXiv:astro-ph/0206308.
- [9] S. R. Furlanetto, *Monthly Notices of the Royal Astronomical Society* **370**, 1867 (2006), arXiv:astro-ph/0604223.
- [10] B. Ciardi, P. Labropoulos, A. Maselli, R. Thomas, and S. Zaroubi, *Monthly Notices of the Royal Astronomical Society* **428**, 1755 (2013), arXiv:1209.2615.
- [11] B. Ciardi, S. Inoue, K. J. Mack, Y. Xu, and G. Bernardi, in *Advancing Astrophysics with the Square Kilometre Array*, *Proceedings of Science* (2015) p. 006, arXiv:1501.04425.
- [12] J. Cang, B. Ciardi, B. Maity, A. J. Nishizawa, Q. Niu, Y. Shao, A. K. Shaw, H. Shimabukuro, T. Soltinský, T.-Y. Sun, T. T. Takeuchi, Y. Xu, K. Yoshikawa, B. Yue, and X. Zhang, *Exploring the cosmic dawn through the 21 cm forest and high-redshift radio sources with the SKA* (2026), arXiv:2606.24665 [astro-ph.CO].
- [13] Z. Haiman, E. Quataert, and G. C. Bower, *The Astrophysical Journal* **612**, 698 (2004), arXiv:astro-ph/0403104.
- [14] E. Bañados, B. P. Venemans, E. Morganson, J. Hodge, R. Decarli, F. Walter, D. Stern, E. Schlafly, E. P. Farina, J. Greiner, K. C. Chambers, X. Fan, H.-W. Rix, W. S. Burgett, P. W. Draper, H. Flewelling, N. Kaiser, N. Metcalfe, J. S. Morgan, J. L. Tonry, and R. J. Wainscoat, *The Astrophysical Journal* **804**, 118 (2015), arXiv:1503.04214 [astro-ph.GA].
- [15] T. Šoltinský, J. S. Bolton, M. Molaro, N. Hatch, M. G. Haehnelt, L. C. Keating, G. Kulkarni, and E. Puchwein, *Monthly Notices of the Royal Astronomical Society* **519**, 3027 (2023).
- [16] P. M. Keller, N. Thyagarajan, A. Kumar, N. Kanekar, and G. Bernardi, *Monthly Notices of the Royal Astronomical Society* **528**, 5692 (2024).
- [17] Q. Niu, Y. Li, Y. Xu, H. Guo, and X. Zhang, *The Astrophysical Journal* **978**, 145 (2025), arXiv:2407.18136 [astro-ph.CO].
- [18] S. Zhang, Q. Niu, Y. Li, and X. Zhang, *Chinese Physics C* **49**, 115105 (2025), arXiv:2504.15086 [astro-ph.CO].
- [19] Y. Xu, X. Chen, Z. Fan, H. Trac, and R. Cen, *The Astrophysical Journal* **704**, 1396 (2009), arXiv:0904.4254.
- [20] Y. Xu, A. Ferrara, and X. Chen, *Monthly Notices of the Royal Astronomical Society* **410**, 2025 (2011), arXiv:1009.1149.
- [21] B. Semelin, *Monthly Notices of the Royal Astronomical Society* **455**, 962 (2016), arXiv:1510.02296.
- [22] T. Šoltinský, A. Chakraborty, G. Kulkarni, M. Viel, C. M. Trott, R. Sagar, N. Thyagarajan, J. S. Bolton, B. Ciardi, E. V. Ryan-Weber, S. Maitra, A. Datta, and N. Roy, *arXiv e-prints*, arXiv:2607.15341 (2026), arXiv:2607.15341 [astro-ph.CO].
- [23] C. Kongprachaya, G. Bernardi, E. Ceccotti, A. Mesinger, B. Ciardi, and O. M. Smirnov, *The burgeoning rise of the 21-cm forest i: constraints on the optical depth of the intergalactic medium at $5.38 < z < 5.84$* (2026), arXiv:2607.20371 [astro-ph.CO].
- [24] CD Science Working Group, G. Bernardi, D. Breitman, A. Datta, A. Fialkov, L. V. E. Koopmans, A. Liu, Y. Mao, G. Mellema, F. Mertens, A. Mesinger, J. Pritchard, A. Schneider, C. M. Trott, Y. Xu, A. Acharya, S. Bag, R. Barkana, N. Barry, O. Basquette, M. Bianco, S. A. Brackenhoff, C. Burigana, I. P. Carucci, E. Ceccotti, A. Chakraborty, S. Choudhuri, T. R. Choudhury, B. Ciardi, H. A. G. Cruz, S. Dasgupta, K. K. Datta, P. Dayal, E. de Lera Acedo, K. M. A. Elahi, I. Georgiev, S. Singh Gill, S. K. Giri, A. Gorce, Q. Guo, C. Heneka, I. Hothi, A. Hutter, P. Kittiwisit, B. Li, Y. Mahida, B. Maity, S. Majumdar, A. Meiksin, R. Meriot, A. Mishra, S. Mittal, R. Mondal, L. Moscardini, J. Munoz, S. Munshi, P. K. Natwariya, L. Noble, O. S. D. O'Hara, P. Ocvirk, S. K. Pal, Y. Pietschke, S. G. Piyarat, R. Sagar, R. M. Samir, M. G. Santos, B. Semelin, R. Shah, Y. Shao, A. K. Shaw, H. Shimabukuro, P. H. Sims, T. Soltinsky, G. Sun, A. Tripathi, T. Trombetti, C. Ulusoy, M. Umberto, J. Weller, S. Yatawatta, S. Yoshiura, B. Yue, S. Zaroubi, and Q. Zheng, *arXiv e-prints*, arXiv:2606.26435 (2026), arXiv:2606.26435 [astro-ph.CO].
- [25] A. Bonaldi, T. L. Bourke, P. Hartley, T. An, M. Auddard, O. Bayandina, N. Bellomo, E. Bianchi, M. Burgay, J. Callingham, S. Camera, V. Casasola, V. Cuciti, P. Cole, N. Gupta, C. L. Hale, I. Harrison, J. Hessels, T. Huege, B. C. Joshi, A. Karastergiou, D. Lal, A. Liu, J. Miller-Jones, S. A. Mao, J. Moldon, L. K. Morabito, K. Mulrey, S. O'Sullivan, D. Oberoi, D. J. Pisano, K. Rajwade, M. T. Sargent, R. Sharma, M. Spinelli, X. Sun, F. S. Tabatabaei, C. Trott, J. T. van Loon, T. Vernstrom, J. Wagg, J. Wang, K. Wang, S. Wedemeyer, J. L. West, P. Woudt, J. Yang, and P. Zucca, *arXiv e-prints*, arXiv:2606.20366 (2026), arXiv:2606.20366 [astro-ph.IM].
- [26] H. Shimabukuro, K. Ichiki, S. Inoue, and S. Yokoyama, *Physical Review D* **90**, 083003 (2014), arXiv:1403.1605.
- [27] H. Shimabukuro, K. Ichiki, and K. Kadota, *Physical Review D* **101**, 043516 (2020), arXiv:1910.06011.
- [28] H. Shimabukuro, K. Ichiki, and K. Kadota, *Physical Review D* **102**, 023522 (2020), arXiv:2005.05589.
- [29] H. Shimabukuro, K. Ichiki, and K. Kadota, *Physical Review D* **107**, 123520 (2023), arXiv:2212.08409.
- [30] P. Villanueva-Domingo and K. Ichiki, *Publications of the Astronomical Society of Japan* **75**, S33 (2023), arXiv:2104.10695.
- [31] M. Kawasaki, W. Nakano, H. Nakatsuka, and

- E. Sonomoto, *Journal of Cosmology and Astroparticle Physics* **2021** (04), 019, arXiv:2010.13504.
- [32] N. Thyagarajan, *The Astrophysical Journal* **899**, 16 (2020), arXiv:2006.10070 [astro-ph.CO].
- [33] T. Soltinský, J. S. Bolton, N. Hatch, M. G. Haehnelt, L. C. Keating, G. Kulkarni, E. Puchwein, J. Chardin, and D. Aubert, *Monthly Notices of the Royal Astronomical Society* **506**, 5818 (2021), arXiv:2105.02250.
- [34] Y. Shao, Y. Xu, Y. Wang, W. Yang, R. Li, X. Zhang, and X. Chen, *Nature Astronomy* **7**, 1116 (2023), arXiv:2307.04130.
- [35] T.-Y. Sun, Y. Shao, Y. Li, Y. Xu, H. Wang, and X. Zhang, *Communications Physics* **8**, 220 (2025), arXiv:2407.14298.
- [36] Y. Shao, T.-Y. Sun, M.-L. Zhao, and X. Zhang, *Physical Review D* **112**, 063513 (2025), arXiv:2411.17094.
- [37] S. Mallat, *Communications on Pure and Applied Mathematics* **65**, 1331 (2012).
- [38] M. Andreux, T. Angles, G. Exarchakis, R. Leonarduzzi, G. Rochette, L. Thiry, J. Zarka, S. Mallat, J. Andén, E. Belilovsky, J. Bruna, V. Lostanlen, M. J. Chaudhary, M. J. Hirn, E. Oyallon, S. Zhang, C. Cella, and M. Eickenberg, *Journal of Machine Learning Research* **21**, 1 (2020), arXiv:1812.11214.
- [39] G. Valogiannis and C. Dvorkin, *Physical Review D* **105**, 103534 (2022), arXiv:2108.07821 [astro-ph.CO].
- [40] G. Valogiannis and C. Dvorkin, *Physical Review D* **106**, 103509 (2022), arXiv:2204.13717 [astro-ph.CO].
- [41] S. Cheng, Y.-S. Ting, B. Ménard, and J. Bruna, *Monthly Notices of the Royal Astronomical Society* **499**, 5902 (2020), arXiv:2006.08561 [astro-ph.CO].
- [42] S. Cheng and B. Ménard, *Monthly Notices of the Royal Astronomical Society* **507**, 1012 (2021), arXiv:2103.09247 [astro-ph.CO].
- [43] S. Cheng, G. A. Marques, D. Grandón, L. Thiele, M. Shirasaki, B. Ménard, and J. Liu, *Journal of Cosmology and Astroparticle Physics* **2025** (01), 006, arXiv:2404.16085 [astro-ph.CO].
- [44] B. Greig, Y.-S. Ting, and A. A. Kaurov, *Monthly Notices of the Royal Astronomical Society* **513**, 1719 (2022), arXiv:2204.02544 [astro-ph.CO].
- [45] I. Hothi, E. Allys, B. Semelin, and F. Boulanger, *Astronomy and Astrophysics* **686**, A212 (2024), arXiv:2311.00036 [astro-ph.CO].
- [46] X. Zhao, S. Zuo, and Y. Mao, *3d ScatterNet: Inference from 21 cm light-cones* (2023), arXiv:2307.09530 [astro-ph.IM].
- [47] H. Shimabukuro, Y. Xu, and Y. Shao, *Physical Review D* **112**, 063557 (2025), arXiv:2504.14656.
- [48] H. Edelsbrunner, D. Letscher, and A. Zomorodian, *Discrete and Computational Geometry* **28**, 511 (2002).
- [49] A. Zomorodian and G. Carlsson, *Discrete and Computational Geometry* **33**, 249 (2005).
- [50] G. Carlsson, *Bulletin of the American Mathematical Society* **46**, 255 (2009).
- [51] R. Ghrist, *Bulletin of the American Mathematical Society* **45**, 61 (2008).
- [52] T. Sousbie, *Monthly Notices of the Royal Astronomical Society* **414**, 350 (2011), arXiv:1009.4015 [astro-ph.CO].
- [53] P. Pranav, H. Edelsbrunner, R. van de Weygaert, G. Vegter, M. Kerber, B. J. T. Jones, and M. Wintraecken, *Monthly Notices of the Royal Astronomical Society* **465**, 4281 (2017), arXiv:1608.04519 [astro-ph.CO].
- [54] W. Elbers and R. van de Weygaert, *Monthly Notices of the Royal Astronomical Society* **520**, 2709 (2023), arXiv:2209.03948 [astro-ph.CO].
- [55] J. H. T. Yip, M. Biagetti, A. Cole, K. Viswanathan, and G. Shiu, *Journal of Cosmology and Astroparticle Physics* **2024** (09), 034, arXiv:2403.13985 [astro-ph.CO].
- [56] H. Shimabukuro, *Physical Review D* **113**, 083525 (2026), arXiv:2511.13092 [astro-ph.CO].
- [57] P. Coles and L.-Y. Chiang, *Nature* **406**, 376 (2000), arXiv:astro-ph/0006017.
- [58] B. Greig, Y.-S. Ting, and A. A. Kaurov, *Monthly Notices of the Royal Astronomical Society* **519**, 5288 (2023), arXiv:2207.09082 [astro-ph.CO].
- [59] E. Giesel, R. Reischke, B. M. Schäfer, and D. Chia, *Journal of Cosmology and Astroparticle Physics* **2021** (01), 005, arXiv:2005.01057 [gr-qc].
- [60] D. Prelogović and A. Mesinger, *Astronomy and Astrophysics* **688**, A199 (2024), arXiv:2401.12277 [astro-ph.CO].
- [61] C. Sui, X. Zhao, T. Jing, and Y. Mao, in *Machine Learning for Astrophysics* (2023) p. 30, arXiv:2307.04994 [astro-ph.CO].
- [62] C. Sui, Y. Mao, X. Zhao, T. Jing, and B. D. Wandelt, *The Astrophysical Journal* **998**, 329 (2026), arXiv:2511.08716 [astro-ph.CO].
- [63] M. Tegmark, A. Taylor, and A. Heavens, *The Astrophysical Journal* **480**, 22 (1997), arXiv:astro-ph/9603021.
- [64] A. Mesinger, S. Furlanetto, and R. Cen, *Monthly Notices of the Royal Astronomical Society* **411**, 955 (2011), arXiv:1003.3878 [astro-ph.CO].
- [65] J. Park, A. Mesinger, B. Greig, and N. Gillet, *Monthly Notices of the Royal Astronomical Society* **484**, 933 (2019), arXiv:1809.08995 [astro-ph.GA].
- [66] S. G. Murray, B. Greig, A. Mesinger, J. B. Muñoz, Y. Qin, J. Park, and C. A. Watkinson, *Journal of Open Source Software* **5**, 2582 (2020), arXiv:2010.15121 [astro-ph.IM].
- [67] J. Theiler, S. Eubank, A. Longtin, B. Galdrikian, and J. D. Farmer, *Physica D: Nonlinear Phenomena* **58**, 77 (1992).
- [68] T. Schreiber and A. Schmitz, *Physica D: Nonlinear Phenomena* **142**, 346 (2000), arXiv:chao-dyn/9909037.
- [69] S. Kullback and R. A. Leibler, *Annals of Mathematical Statistics* **22**, 79 (1951).
- [70] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. (Wiley, Hoboken, New Jersey, 2006).
- [71] R. A. Fisher, *Mathematical Proceedings of the Cambridge Philosophical Society* **22**, 700 (1925).
- [72] C. R. Rao, *Bulletin of the Calcutta Mathematical Society* **37**, 81 (1945).
- [73] S.-i. Amari, *Information Geometry and Its Applications* (Springer, Tokyo, 2016).
- [74] W. R. Coulton and B. D. Wandelt, arXiv e-prints, arXiv:2305.08994 (2023), arXiv:2305.08994 [stat.ME].
- [75] H. Jeffreys, *Proceedings of the Royal Society of London. Series A* **186**, 453 (1946).
- [76] J. Schäfer and K. Strimmer, *Statistical Applications in Genetics and Molecular Biology* **4**, Article 32 (2005).
- [77] O. Ledoit and M. Wolf, *Journal of Multivariate Analysis* **88**, 365 (2004).
- [78] T. T. Takeuchi, K. Yata, K. Egashira, M. Aoshima, A. Ishii, S. Cooray, K. Nakanishi, K. Kohno, and K. T. Kono, *The Astrophysical Journal Supplement Series* **271**, 44 (2024), arXiv:2203.04535 [astro-ph.IM].